Construction of invariants in knot theory

An introduction to research in mathematics (T.I.P.E.)

Arthur Filippi

arthur.filippi@polytechnique.edu

Lycée Sainte-Geneviève (CPGE), Versailles

Collective work, carried out with Maxime Deporte, Adam Guédiche and David James Lodzinsky.

2023

Abstract

This report is an introductory research project (*T.I.P.E.*, French preparatory classes) on the construction and comparison of knot invariants. After formalising knots as ambient-isotopy classes and reducing their study to diagrams via Reidemeister moves, we build a graded family of invariants of increasing strength: tricolourability; the polynomial invariants of Alexander and Conway, then the Kauffman bracket, the Jones polynomial and the HOMFLY-PT polynomial obtained through skein relations; and the knot group, derived from the fundamental group of the knot complement (Wirtinger presentation). We close with an opening on Vassiliev finite-type invariants and on algorithmic aspects (Gauss codes). Throughout, invariance is established by examining the behaviour of each construction under the (oriented) Reidemeister moves, and the relative power and limitations of the invariants are discussed.

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NOTATION USED

$\mathfrak{K}$	Set of knots
$\mathfrak{L}$	Set of links
$\mathfrak{B}_n$	Braid group on n strands
$R1, R2, ...$	Reidemeister moves
$\mathcal{S}^n$	Hypersphere of dimension n
$\mathbb{T}^2$	Simple torus (i.e. with one hole)
A	Unknot
T_1, T_2, H	Left and right trefoil knots and figure-eight knot
$\mathcal{K}_p$, $\mathcal{K}_m$	Square knot and granny knot
$\mathbb{E}^{(I)}$	Almost-null families of $\mathbb{E}$ indexed by I
$\Delta_K(X)$	Alexander polynomial of the knot K
$\nabla_L(X)$	Conway polynomial of the link L
$\langle L \rangle$	Kauffman bracket of the link L
$J_L(X)$	Jones polynomial of the link L
$P_L(v, z)$	HOMFLY-PT polynomial of the link L
$\mathcal{L}(\Omega, x)$	Set of loops of Ω based at x
$\mathcal{C}(\Omega, x_0, x_1)$	Set of paths of Ω with endpoints x_0 and x_1
$\pi_1(\Omega, x)$	Fundamental group of Ω at x
Π_1	Covariant functor $Top_* \longrightarrow Grp$
ϕ_*	Morphism induced by the map ϕ through the functor Π_1
$G(K)$	Knot group of the knot K
$\mathfrak{S}_n$	Symmetric group of order n
Σ_n	Set of n-singular knots
$\mathfrak{F}$	Set of singular and non-singular knots
$\mathfrak{V}_n$	Space of Vassiliev invariants of order $\leq n$

INTRODUCTION AND AIM

"Imitate as little as possible men in their enigmatic sickness of making knots", urges René Char in his collection *Rougeur des matinaux* (1972): here is a rather bold injunction, to which the mathematician, whether he laughs at it or takes offence, certainly does not remain indifferent. A sickness, says René Char? A journey of a few days among the Caribou Inuit of the central Canadian Arctic would surely change his mind: there he would discover, as the ethnologist Jean Gabus did 30 years before him, the beauty of string games. Not an evening goes by without the Paallirmiut Inuit competing in ingenuity to "produce a great number of figures: the grizzly bear, the polar bear, the musk ox ...". In these lands of Hudson Bay, knots carry a connotation that is as playful as it is educational: they lay claim to a certain *skholè*. The old Inuit sages, past masters in the art of tying and untying strings, have their emulators among mathematicians, who have made this skilful game their own. We offer our reader an overview of knot theory: he will see that this fertile theory draws, in many respects, its inspiration and its motivations from a certain playful imagination. How to obtain, starting from the knot of the *tuktu* (caribou), the figure of the *amarok* (wolf): that is the aim of our work.

1 FORMALISING THE CONCEPT OF A KNOT

The concept of a knot is rather intuitive: take a "piece of string", tangle its ends, then glue them back together. We adopt the following definition:

Definition 1.1 (Knot): A knot K in $\mathbb{R}^3$ is the image of $\mathcal{S}^1$ under a continuous and injective map γ. *We consider here "smooth" knots, that is, we take* $\gamma \in \mathcal{C}^\infty(\mathbb{R}^3, \mathbb{R}^3)$. *We could also have considered "piecewise linear" knots: this does not fundamentally change the problem.*

Example 1.2: The unknot, written A, is $A = \mathcal{S}^1 \subset \mathbb{C} \times \mathbb{R} \cong \mathbb{R}^3$

Definition 1.3 (Link): A link L in $\mathbb{R}^3$ with n components is the image of the disjoint union of n circles $\mathcal{S}^1$ under a continuous and injective map γ. *In the same spirit as for knots, we shall consider smooth links here.*

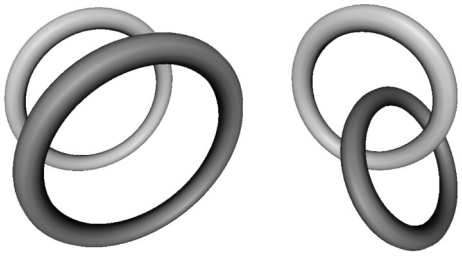


Figure 1: Trivial two-component link (left) and Hopf link (right).

The aim of knot theory is to identify equal knots, that is, those that can be obtained from one another by a sequence of admissible moves, "without tearing the string" and without the string "passing through itself". This is why we introduce the concept of ambient isotopy.

Definition 1.4 (Ambient isotopy): We say that the manifold K' arises from the manifold K by ambient isotopy if there exists a continuous map

$$\begin{array}{rccc} \Phi & : & [0,1]\times\mathbb{R}^3 & \longrightarrow & \mathbb{R}^3 \\ & & (t,x) & \longmapsto & \Phi(t,x)=\Phi_t(x) \end{array}$$

such that:

i) $\Phi_0 = Id_{\mathbb{R}^3}$

ii) $\Phi_1(K) = K'$

iii) $\forall t \in [0,1], \Phi_t$ is a homeomorphism.

Formally, an ambient isotopy is a continuous deformation of space over a time interval, such that at the initial instant space is not yet deformed. Why do we consider a deformation of the whole space rather than of the manifold alone (here, the knot)? It is in order to take into account "the air between the strands", so as to prevent the knot from retracting onto the unknot.

Property 1.5: For two knots K and K', the relation *K' arises from K by ambient isotopy* defines an equivalence relation.

Proof of property 1.5: Let K, K', K" be three knots,

i) Reflexivity: it suffices to consider $\forall t \in [0,1], \Phi_t := Id_{\mathbb{R}^3}$ which indeed gives $K \sim K$

ii) Symmetry: suppose that $K \sim K'$ by the ambient isotopy Φ. It suffices to consider:

$$\begin{array}{rccc} \widetilde{\Phi} & : & [0,1]\times\mathbb{R}^3 & \longrightarrow & \mathbb{R}^3 \\ & & (t,x) & \longmapsto & \widetilde{\Phi}(t,x)=\Phi_{1-t}(x) \end{array}$$

which indeed provides an ambient isotopy from K' to K, by inheritance of the properties of Φ.

iii) Transitivity: suppose that $K \sim K'$ and $K' \sim K''$ by the ambient isotopies Φ and Ψ respectively. We may consider $\Omega := \Psi \circ \Phi$. We do have $\Omega_0 = \Psi_0 \circ \Phi_0 = Id_{\mathbb{R}^3}$, $\Omega_1(K) = \Psi_1(\Phi_1(K)) = \Psi_1(K') = K$", and finally, by composition of homeomorphisms, $\forall t \in [0,1], \Omega_t = \Psi_t \circ \Phi_t$ is a homeomorphism, whence $K \sim K''$.

□

From now on we shall therefore, with a slight abuse of language, consider a knot as an isotopy class. This is the fundamental idea of knot theory: to pass from one knot to another by an isotopy.

To simplify our approach, we shall consider "knot diagrams", which are, in a way, the projection of the knot onto a plane.

Definition 1.6 (Knot diagram): A knot diagram is the *generic* immersion of a knot K of $\mathbb{R}^3$ into $\mathbb{R}^2 \times \{0, \epsilon\}$. The term "generic" means that the diagram is explicit enough for there to be no ambiguity, in the sense that any multiple points are only double points with distinct tangents. The "over-under" information (necessary when projecting) is contained in the coordinate along $\{0, \epsilon\}$. From now on we shall work on knot diagrams.

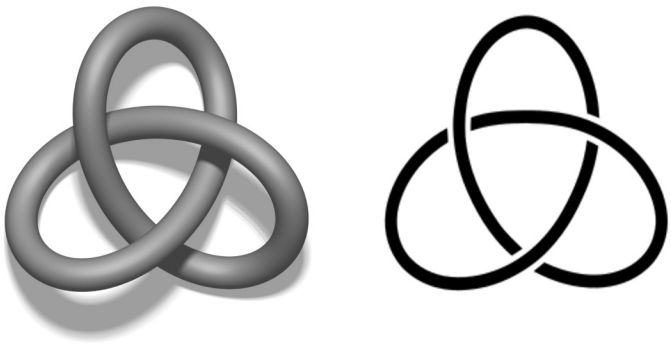


Figure 2: Left trefoil knot T_1 and its diagram.

Definition 1.7 (Isotopy of diagrams): We say that the diagram K' arises from the diagram K by isotopy if there exists a continuous map

$$\begin{array}{rccc} \Psi & : & [0,1] \times K & \longrightarrow & \mathbb{R}^2 \times \{0, \epsilon\} \\ & & (t, x) & \longmapsto & \Psi(t, x) = \Psi_t(x) \end{array}$$

such that:

i) $\Psi_0 = Id_K$

ii) $\Psi_1(K) = K'$

iii) $\forall t \in [0,1], \Psi_t$ is a homeomorphism

Proposition 1.8 (stated without proof): Every diagram represents a unique knot (up to isotopy).

Proposition 1.9 (stated without proof): Two isotopic diagrams represent the same knot. We shall then allow ourselves to speak of *the* diagram of a knot, and more generally to identify a knot with its diagram.

2 CONSTRUCTION OF INVARIANTS

2.1 HISTORICAL ATTEMPTS

The aim is to classify knots up to isotopy. To that end, we define the notion of an invariant.

Definition 2.1.1 (Knot invariant): We write $\mathfrak{K}$ for the set of knots. Let $\mathbb{E}$ be a set (of numbers, of functions, of groups, of topological spaces, etc...). We say that $\Gamma : \mathfrak{K} \longrightarrow \mathbb{E}$ is a knot invariant when $\forall K, K' \in \mathfrak{K}, [K \sim K' \Rightarrow \Gamma(K) = \Gamma(K')]$. Short of being able to tell when two knots are isotopic, Γ makes it possible to detect two knots that *are not* isotopic (by contraposition).

Definition 2.1.2 (Complete knot invariant): Let us keep the notation of definition 2.1. We shall say that Γ is complete when: $\forall K, K' \in \mathfrak{K}, [K \sim K' \Leftrightarrow \Gamma(K) = \Gamma(K')]$. Such an invariant makes it possible to identify when two knots *are* isotopic (which is much stronger).

Several historical attempts have failed to construct, if not invariants, at least tools for identifying knots. We mention them here briefly for reference: they also show how fertile knot theory is and how many perspectives it requires.

2.1.1 BRAID GROUP, BRAID CLOSURES

A braid corresponds to the common idea of intertwined threads: a braid of hair, "scoubidou" bracelets, and so on. Here, for example, are braids on 3 and 5 strands respectively:

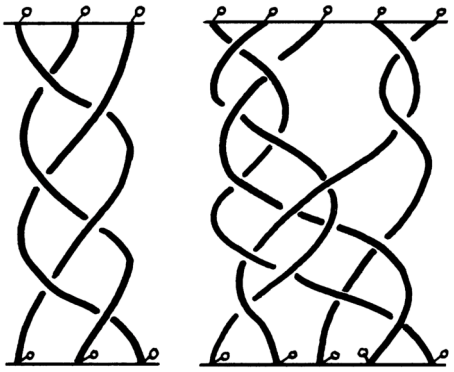
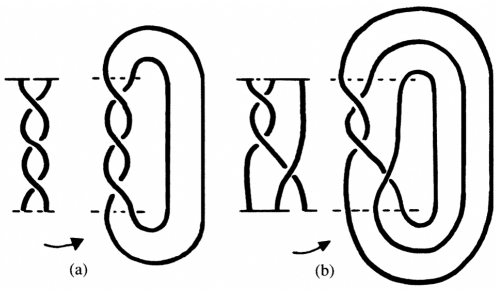


Figure 3: Braids on 3 and 4 strands respectively.

The closure operation makes it possible to obtain a link, and sometimes a knot, from a braid, as shown in figure 4.1. Figure 4.2 shows the inverse operation.



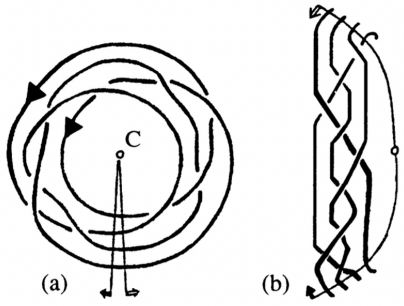


Figure 4.1 (left): Braid closures, obtaining a knot (a) and a link (b).
Figure 4.2 (right): Unwinding a knot into a braid.

Theorem 2.1.3 (Alexander) (stated without proof): Every knot is obtained as the closure of some braid.

The isotopy relation extends to braid diagrams. Equipped with the product operation (attaching the upper braids to the lower ones), the set $\mathfrak{B}_n$ of braids on n strands is a group (non-abelian as soon as $n > 2$). (One easily checks the existence of an identity braid and the invertibility of every braid.) Moreover $\mathfrak{B}_n$ is generated by $n-1$ elementary braids, which makes it possible to replace a braid by its "braid word". An example is given in figure 5.

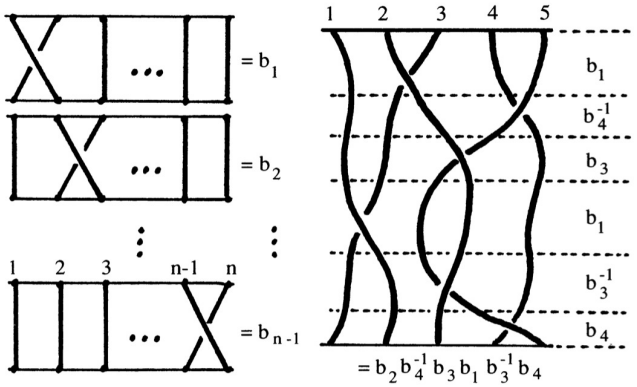


Figure 5: Generators of $\mathfrak{B}_n$ and the word of a braid.

We have the following two relations (admissible computations):

i) Commutativity of distant braids: $\forall (i,j) \in \{1,2,...,n-1\}, [|i-j| \geq 2 \Rightarrow b_i b_j = b_j b_i]$

ii) Artin relation: $\forall i \in \{1,2,...,n-2\}, b_i b_{i+1} b_i = b_{i+1} b_i b_{i+1}$

Theorem 2.1.4 (Artin, 1925) (stated without proof): Two braids are isotopic (in the sense of definition 1.3) if, and only if, the word of one can be obtained from the word of the other by a sequence of admissible computations.

Nevertheless it has still not been possible to classify knots by way of the braid group.

2.1.2 ARITHMETIC OF KNOTS

For this reason, another approach was considered: the arithmetic of knots. The ambition was to succeed where the braid group had failed. To do so, knots had to be equipped with a structure on which it is possible to reason. It is with this in mind that one defines an operation of "composition" of knots similar to the composition of braids:

Definition 2.1.5 (Connected sum): Let J, K be two knots. Consider $D_{J \sqcup K}$ the diagram of their disjoint union. It is immediate that one can find a disc in space resembling the left-hand part of figure 6. Then the connected sum $J \# K$ is the knot corresponding to the diagram $D_{J \sqcup K}$, with the difference that the disc on the left of figure 6 has been replaced by the one on the right.
We take for granted that the connected sum of J and K is independent of the chosen disc.

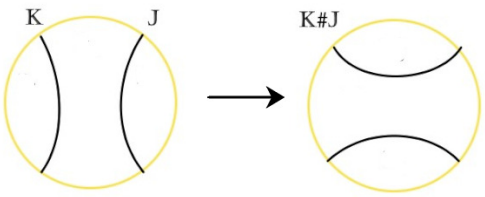


Figure 6: Realisation of the connected sum of $J \sqcup K$

Property 2.1.6: The connected sum is associative, commutative and admits the unknot A as an identity element.

Proof of property 2.1.6: Associativity is immediate from the definition. Moreover, one checks just as easily that A works as an identity element. Finally, commutativity can be visualised by "sliding" J inside K.

□

These properties give $(\mathfrak{K}, \#)$ the structure of a commutative monoid.

Definition 2.1.7: A knot K is said to be composite if it can be written as the connected sum of two non-trivial knots J and J', i.e. when there exist two non-trivial knots J, J' such that $K = J \# J'$.
A knot is said to be prime if it is neither trivial nor composite.

Armed with these definitions, it is at last possible to classify knots thanks to the:

Theorem 2.1.8 (Schubert, 1940s) (stated without proof): Every knot decomposes in a unique way (up to the order of the factors) as a connected sum of prime knots.

Unfortunately, this arithmetic approach, entirely complete in theoretical terms, fails in practice. Indeed, it is very complicated to establish the prime knots algorithmically, and to give the decomposition into prime knots of an arbitrary knot: it is, so to speak, a genuine Gordian knot...

2.2 REIDEMEISTER MOVES

The aim of this paragraph is to find a way of easily checking that a function on knots is invariant under isotopy, and therefore that it defines a knot invariant.

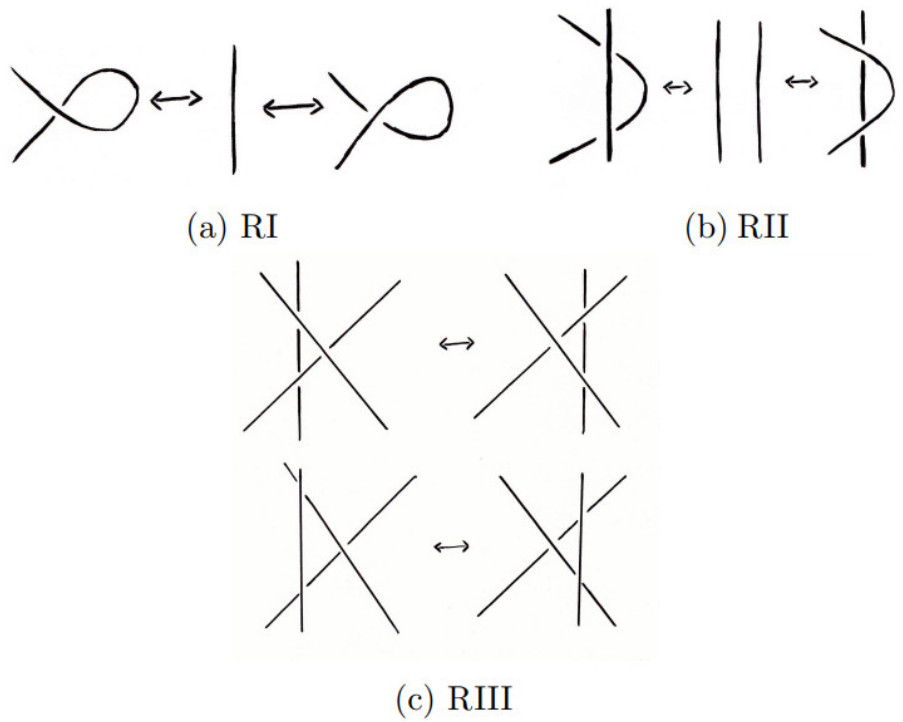


Figure 7: The three Reidemeister moves.

Theorem 2.2.2 (Reidemeister, 1927) (stated without proof): Two knot diagrams are isotopic if and only if one can pass from one to the other by a finite sequence of the Reidemeister moves and their inverses shown in figure 7.

Remark 2.2.3: Strictly speaking, one would have to consider a fourth "trivial" type of move, written R0, corresponding to planar isotopies, that is, the continuous deformations that do not modify the relative positions of the crossings of the diagram.

Thanks to this theorem, in order to show that a given function is indeed a knot invariant, it then suffices to check that it is invariant under the three Reidemeister moves.

2.3 A FIRST INVARIANT: TRICOLOURABILITY

Definition 2.3.1: A knot diagram is said to be *tricolourable* when it can be coloured with three different colours according to the following rules:

i) Each strand of the knot (*i.e.* each portion of the knot lying between two crossings) is coloured with a single colour

ii) Each of the three colours is used to colour the knot

iii) At each crossing, one must find either a single colour, or all three colours

iv) A change of colour can only take place at a hidden point of the cord (i.e. at a crossing, on the string that passes underneath the crossing)

Proposition 2.3.2: Tricolourability is a knot invariant.

Proof of proposition 2.3.2: it suffices to examine the effect of the three Reidemeister moves (and of their inverses) on the tricolourability of a knot diagram. To be precise, we want to show that applying a Reidemeister move to a tricolourable knot preserves its tricolourability, i.e. that it remains possible to find a tricolouring. In what follows we are given a tricolouring of a diagram and we describe how to deduce from it a tricolouring of the diagram obtained by each Reidemeister move, and conversely.

- For $R1$, rule 3 forces the loop to be of a single colour. One can then obtain a tricolouring of the diagram obtained by $R1$ simply by keeping the same colour:

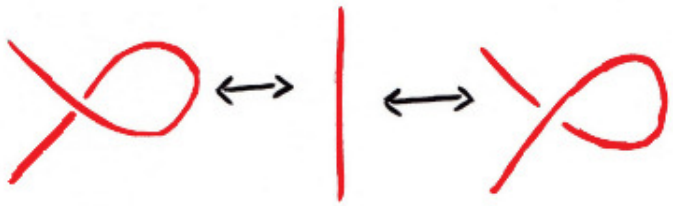


Figure 8: Invariance under $R1$.

- For $R2$, up to a permutation of the colours, rule 3 leaves only 2 possibilities, which we treat as in the following diagrams:

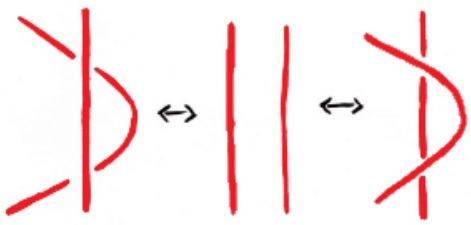


Figure 9: Invariance under $R2$, first possibility.

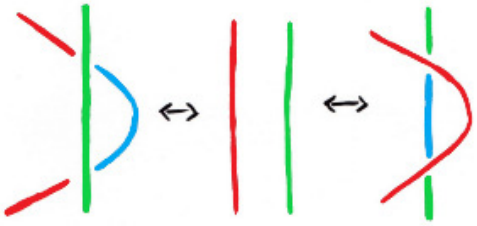


Figure 10: Invariance under $R2$, second possibility.

In both cases, we observe that the rest of the diagram continues to satisfy rules 1, 2 and 4.

- For R3, we note that the choice of the colours of the 3 darkened strands on the first diagram of the following figure fixes the colours of the other strands. Up to a permutation of the colours, only 5 possibilities then remain to be examined.

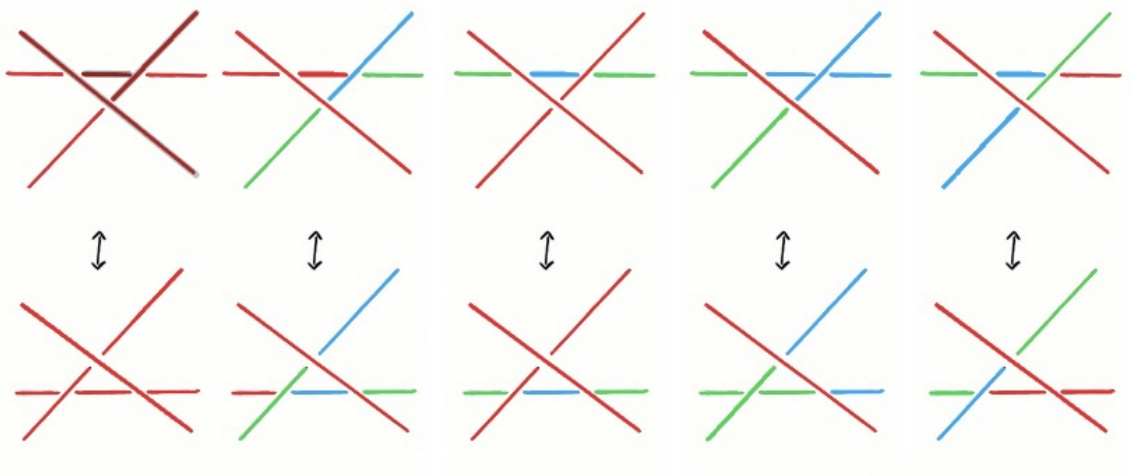


Figure 11: All the possibilities for $R3$.

□

Example 2.3.3: The trefoil knot cannot be untied: it is not isotopic to the unknot. Indeed, the trefoil knot is tricolourable, whereas the unknot clearly is not:



Figure 12: The tricoloured trefoil knot.

2.4 POLYNOMIAL INVARIANTS

Polynomial invariants may appear more general than knot invariants because, for the most part, their purpose is to make it possible to distinguish two links from one another. A knot being a link with 1 component, this would *a priori* allow one to check whether two knots are isotopic.

2.4.1 PRELIMINARIES

In order to meet this new requirement, we extend the definitions of knot invariant, complete knot invariant and knot diagram in the following way:

Definition 2.4.1 (Link invariant): We write $\mathfrak{L}$ for the set of links. Let $\mathbb{E}$ be a set (of numbers, of groups, of topological spaces, etc...). We say that $\Gamma : \mathfrak{L} \longrightarrow \mathbb{E}$ is a link invariant when $\forall L, L' \in \mathfrak{L},\ [L \sim L' \Rightarrow \Gamma(L) = \Gamma(L')]$. In the same way as knot invariants, link invariants make it possible at most to check that two knots are *not* isotopic.

Definition 2.4.2 (Complete link invariant): Let us keep the notation of definition 2.4.1.1. We shall say that Γ is complete when: $\forall L, L' \in \mathfrak{L},\ [L \sim L' \Leftrightarrow \Gamma(L) = \Gamma(L')]$. Such an invariant, for its part, makes it possible to identify when two links *are* isotopic (which is much stronger).

Definition 2.4.3 (Link diagram): A link diagram is the *generic* immersion of a link L of $\mathbb{R}^3$ into $\mathbb{R}^2 \times \{0, \epsilon\}$. The term "generic" means that the diagram is explicit enough for there to be no ambiguity, in the sense that any multiple points are only double points with distinct tangents. The "over-under" information (necessary when projecting) is contained in the coordinate along $\{0, \epsilon\}$.

Remark 2.4.4: Propositions 1.8 and 1.9 remain true for links: every diagram represents a unique link (up to isotopy) and two isotopic diagrams represent the same link. We shall then allow ourselves to speak of *the* diagram of a link, and more generally to identify a link with its diagram.
Moreover, the result of Reidemeister's theorem extends naturally to links: it suffices to be invariant under the 3 Reidemeister moves in order to be a link invariant.

We now refine the representation of a link with a new parameter: direction.

Definition 2.4.5 (Orientation of a link diagram): Intuitively, we equip the diagram with a direction of travel.

Definition 2.4.6 (Sign of a crossing): Let L be an oriented link with n crossings and D its diagram. We number its crossings in the form $c_1, ..., c_n$. Let $i \in [\![1, n]\!]$. We then define the sign of c_i in the following way:

- If, in order to align the "over-strand" with the "under-strand", one has to turn anticlockwise, then c_i is positive.
- Otherwise, one has to turn clockwise: c_i is negative.

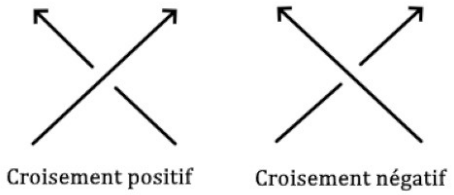


Figure 13: The 2 crossings.

We shall often represent these crossings inside discs. The disc is to be seen as a zoom on a part of the link under consideration: only the part inside the disc will be of interest to us (we shall for instance apply Reidemeister moves to it).

Remark 2.4.7: Isotopy of oriented links is a stronger property than isotopy of links. For example, here are two oriented diagrams that are not isotopic if their orientation is taken into account, whereas the unoriented links are:



Figure 14: 2 orientations of the same knot.

The orientation of a link, which will be useful later on to define certain invariants, unfortunately makes Reidemeister's theorem insufficient: from the 3 moves studied in the previous paragraph, one obtains others by taking the orientation of the link into account. Consequently, a substantial number of checks is required in order to determine the invariance of a function. For example, the 1st move now has 4 variants:

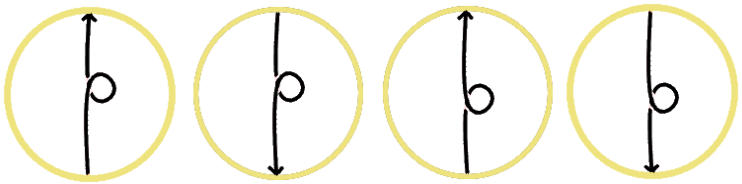


Figure 15: The 4 variants of the 1st Reidemeister move: $R1a$, $R1b$, $R1c$ and $R1d$.

Fortunately, a theorem makes it possible to restrict to only 4 of these moves, which will reduce the number of checks to be carried out in order to prove the invariants:

Theorem 2.4.8 (Polyak-Suwara) (stated without proof): All the oriented Reidemeister moves are generated by the following 4 moves:

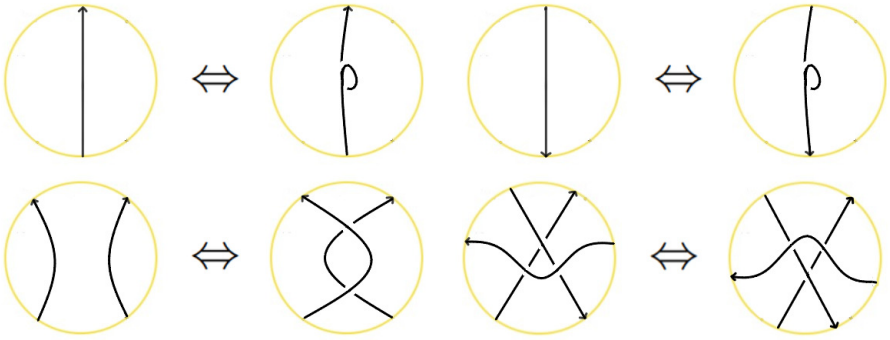


Figure 16: The 4 elementary moves: $R1a$, $R1b$, $R2a$ and $R3a$.

Property 2.4.9 (stated without proof): A link diagram with n crossings cuts the plane into $n+2$ regions diffeomorphic to a disc.

One can intuitively understand this property by identifying a link diagram with a graph in the plane (the

vertices represent the crossings).
Indeed, by an Euler formula the number f of regions then satisfies $f + n - m = 2$ where n is the number of vertices and m the number of edges of the graph.
Now, each vertex being of course of degree 4, we have $m = 2n$ and then $f = n + 2$.

Definition 2.4.10 (Laurent polynomial): A Laurent polynomial is a generalisation of polynomials to negative powers. We define a polynomial P as an almost-null family $P = (a_k)_{k\in\mathbb{Z}} \in \mathbb{A}^{(\mathbb{Z})}$ where $\mathbb{A}$ is a commutative ring. In this report we shall be interested in particular in $\mathbb{A} = \mathbb{Z}$.
One can then write P in the form: $P = \sum_{k\in\mathbb{Z}} a_k X^k$, where X is the indeterminate.
We shall write the set of Laurent polynomials over the ring $\mathbb{A}$ in the following way: $\mathbb{A}[X, X^{-1}]$.

2.4.2 ALEXANDER POLYNOMIAL

James Waddell Alexander II was the first to take an interest in polynomial invariants. He constructed in particular one that would bear his name.

Several definitions of the Alexander polynomial exist. We shall first be interested in the first historical definition, then in the one that is the most practical to use.

Definition 2.4.11: Let K be a knot with n crossings and D its diagram. We define a matrix $A \in \mathcal{M}_{n,n+2}$ as follows:
We arbitrarily orient D, then index its crossings in the form $c_1, ..., c_n$ and the regions of the plane that it delimits in the form $R_0, ..., R_{n+1}$ (R_0 will be the unbounded region).
The rows of A correspond to the crossings and the columns to the regions.
We consider a crossing c_i with $i \in [\![1, n]\!]$ and its under-strand. We travel through the crossing in the direction given by the orientation of the knot and we set $A_{i,j} :=$

- $-X$ if R_{j-1} is the region to the left of the strand before the crossing.
- 1 if R_{j-1} is the region to the right of the strand before the crossing.
- X if R_{j-1} is the region to the left of the strand after the crossing.
- -1 if R_{j-1} is the region to the right of the strand after the crossing.
- 0 if the region R_{j-1} is not adjacent to c_i.

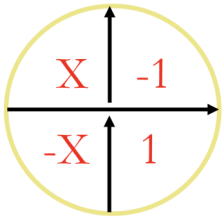


Figure 17: Coefficients around a crossing.

Note that if, according to the previous definition, $A_{i,j}$ has to take two values simultaneously, then it will be equal to the sum of these values.
We then remove 2 columns of A corresponding to adjacent regions of the plane (which share a strand of the knot as a boundary) in order to obtain a matrix B $\in \mathcal{M}_{n,n}(\mathbb{Z}[X, X^{-1}])$.

Definition 2.4.12 (Alexander polynomial): The Alexander polynomial of the knot K is defined as $\Delta_K(X) = [det(B)] \in \mathbb{Z}[X, X^{-1}]/\{\pm X^{\mathbb{Z}}\}$.
Intuitively, it is the equivalence class of the determinant of B up to a factor X^k ($k \in \mathbb{Z}$).
We shall write indifferently $\Delta_K(X)$, $\Delta(A)$ and $\Delta(B)$ where A and B are the matrices defined as above. Indeed, we shall show shortly that the various matrices A and B that can be obtained from K all lead to the same Alexander polynomial.

Example 2.4.13: The example of the right trefoil knot T_2.

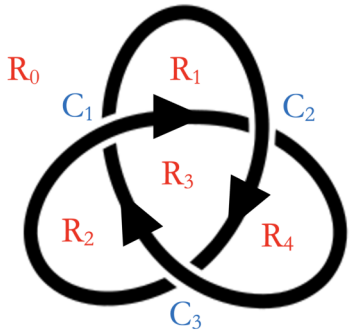


Figure 18: Oriented right trefoil knot T_2.

From this diagram, we determine the following matrix A:

$$A = \begin{pmatrix} X & -1 & -X & 1 & 0 \\ X & -X & 0 & 1 & -1 \\ X & 0 & -1 & 1 & -X \end{pmatrix}$$

To determine B, we shall delete 2 regions. Here, we choose to delete R_0 and R_1, which correspond to the first 2 columns. For readers who question this arbitrary choice, we shall show in what follows that the polynomial does not depend on it.
It follows that: $\Delta_K(X) = det(B) = X^2 - X + 1$

Property 2.4.14: The Alexander polynomial is multiplicative for the connected sum.
Formally, let K and J be two knots. Then $\Delta_{K\#J}(X) = \Delta_K(X)\Delta_J(X)$.

Proof of property 2.4.14: Let K,J be two knots. Write D_K and D_J for their respective diagrams. Then set D to be the diagram of K#J. Next write $c_1, ..., c_n$ and $R_0, ..., R_{n+1}$ for the regions of D_K and $c'_1, ..., c'_p$ and $R'_0, ..., R'_{p+1}$ for the regions of D_j.
Figure 18 confirms that D has crossings $c_1, ..., c_n, c'_1, ..., c'_p$ (they are not modified by the connected sum) and regions $R''_0, R''_1, R_2, ..., R_{n+1}, R'_2, ..., R'_{p+1}$ (because R_1 and R'_1 become the same zone R''_1 after the connected sum).

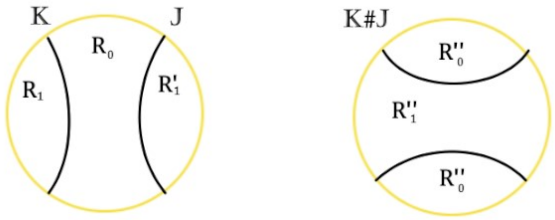


Figure 19: Disc of change

Then we write $A(D_K)_{R_0,R_1}$ for the matrix of D_k from which the columns R_0 and R_1 have been removed. As D differs from the diagrams of the knots only at R_0'' and R_1'', $A(D)_{R_0'',R_{n+1}''}$ is exactly equal to:

$$A(D)_{R_0'',R_1''} = \begin{pmatrix} A(D_K)_{R_0,R_1} & (0) \\ (0) & A(D_J)_{R_0,R_1'} \end{pmatrix} \in \mathcal{M}_{n+p}(\mathbb{Z}[X, X^{-1}])$$

By a block triangular matrix argument, we obtain the desired result.

□

Theorem 2.4.15: The Alexander polynomial does not depend on the choice:

- Of the order of the regions.
- Of the order of the crossings.
- Of the pair of columns removed.

Proof of theorem 2.4.15: Invariance under the choice of the order of the regions (respectively of the crossings) is immediate because such a choice corresponds to a permutation of the columns (respectively of the rows) of the matrix, which makes its determinant vary by a factor $\pm X^0$.

Invariance under the choice of the pair of columns removed is a little more delicate to show.
We shall first introduce the "number" $n(R)$ of a region R.
Let K be an oriented knot with n crossings, whose regions and crossings are indexed as before.
Let $A(K) \in (M_{n,n+2})$ be the matrix of K. Let $p \in \mathbb{Z}$. We set $n(R_0) = p$.
We then determine the "numbers" of the other regions step by step: let R_k and R_j be adjacent with $n(R_k)$ known. We then set $n(R_j) =$

- $n(R_k) + 1$ if, in going from R_k to R_j, we crossed a strand of the knot from right to left (the direction of travel follows from the orientation of the knot).
- $n(R_k) - 1$ if, in going from R_k to R_j, we crossed a strand of the knot from left to right.

We take for granted that such a numbering exists and one can then easily convince oneself of its uniqueness: every region can be reached step by step starting from R_0, which determines the "number" of each region. Here is an example for the trefoil knot:

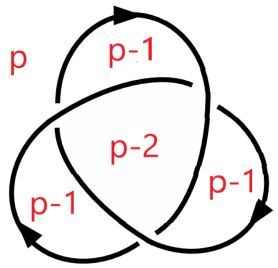


Figure 20: Numbering of the trefoil knot.

Let us first note that, writing S_k for the sum of the columns of A corresponding to the regions satisfying $n(R) = k$, we have $\sum_k S_k = \begin{pmatrix} 0 \\ . \\ . \\ . \\ 0 \end{pmatrix}$ (E_1)

Indeed, $\{R_i \mid i \in [\![0, n+1]\!]\} = \bigsqcup_k E_k$, where E_k denotes the set of R such that $n(R) = k$, then $\sum_k S_k = \sum_i C_i$ and, as the sum of the coefficients of each row of A is zero, $\sum_i C_i = \begin{pmatrix} 0 \\ . \\ . \\ . \\ 0 \end{pmatrix}$

Let us multiply each column of A by X^{-k} (thus defining a new matrix A'), where $k = n(R)$ and R is the region associated with the column. Let us show that the sum of the coefficients of a row of A' is zero: set $k = n(R)$ where R is the region with coefficient X in the row of A.

- If the row corresponds to a positive crossing, then the non-zero coefficients of the row of A' are $X \times X^{-k}, -X \times X^{-(k-1)}, 1 \times X^{-(k-2)}$ and $-1 \times X^{-(k-1)}$.
  Their sum is zero.
- If the row corresponds to a negative crossing, then the non-zero coefficients of the row of A' are $X \times X^{-k}, -X \times X^{-(k+1)}, 1 \times X^{-k}$ and $-1 \times X^{-(k-1)}$.
  Their sum is zero as well.

Thus, as before, we have $\sum_k X^{-k} S_k = \begin{pmatrix} 0 \\ . \\ . \\ . \\ 0 \end{pmatrix}$ (E_1)

Then $\forall n \in \mathbb{Z}$, $\sum_k X^{n-k} S_k = \begin{pmatrix} 0 \\ . \\ . \\ . \\ 0 \end{pmatrix}$ (E_2)

$(E_2) - (E_1)$ then gives $\sum_k (X^{n-k} - 1) S_k = \begin{pmatrix} 0 \\ . \\ . \\ . \\ 0 \end{pmatrix}$

Thus, let C_i be a column of A', whose region has a "number" n_i. We have $(X^{n-k_i} - 1) C_i = \sum_{a \neq i} (X^{n-n_a} - 1) C_a$ where n_a is the "number" of the region associated with C_a.

Armed with this result, we shall be able to move on to the computation of the determinants.
We shall write $\Delta_{i,j}$ for the determinant of the matrix obtained by removing the columns i and j from A.
Let i, j, k, l be pairwise distinct between 0 and $n+1$, and let us show that $\Delta_{i,j}$ and $\Delta_{k,l}$ are equal up to a factor $\pm X^m$.
Let us number the regions of the knot as before and set r, p, q, s to be the respective "numbers" of the regions associated with the columns i, j, k, l.
We have:

$$\frac{X^{r-q}-1}{X^{r-p}-1}\Delta_{i,j} = \frac{X^{r-q}-1}{X^{r-p}-1} \times det \begin{pmatrix} C_0 & ... & C_{i-1} & C_{i+1} & ... & C_{j-1} & C_{j+1} & ... & C_{n+1} \end{pmatrix}$$

$$= det \begin{pmatrix} C_0 & ... & C_{i-1} & C_{i+1} & ... & C_{j-1} & C_{j+1} & ... & \frac{X^{r-q}-1}{X^{r-p}-1} C_k & ... & C_{n+1} \end{pmatrix}$$

$$= det \begin{pmatrix} C_0 & ... & C_{i-1} & C_{i+1} & ... & C_{j-1} & C_{j+1} & ... & \frac{X^{r-q}-1}{X^{r-p}-1} \frac{\sum_{a \neq k} (X^{r-n_a}-1) C_a}{X^{r-q}-1} & ... & C_{n+1} \end{pmatrix}$$

$$= det \begin{pmatrix} C_0 & ... & C_{i-1} & C_{i+1} & ... & C_{j-1} & C_{j+1} & ... & \frac{X^{r-q}-1}{X^{r-p}-1} \frac{\sum_{a \in \{i,j\}} (X^{r-n_a}-1) C_a}{X^{r-q}-1} & ... & C_{n+1} \end{pmatrix}$$

$$= det \begin{pmatrix} C_0 & ... & C_{i-1} & C_{i+1} & ... & C_{j-1} & C_{j+1} & ... & C_j & ... & C_{n+1} \end{pmatrix}$$
$$= \pm \Delta_{i,k}$$

Similarly, $\frac{X^{q-s}-1}{X^{q-r}-1}\Delta_{k,i} = \pm \Delta_{k,l}$
Here we have treated the computation of the determinant in the case $i \leq j \leq k$: the proof would be the same in the other cases.
We therefore obtain $\Delta_{i,j} = \pm \frac{(X^{r-p}-1)(X^{q-r}-1)}{(X^{r-q}-1)(X^{q-s}-1)} \Delta_{k,l}$
Now, $(X^{r-q}-1)X^{q-r} = 1 - X^{q-r}$, i.e. $\frac{X^{q-r}-1}{X^{r-q}-1} = -X^{q-r}$
Finally, $\Delta_{i,j} = \pm \frac{(X^{r-p}-1)X^{q-r}}{X^{q-s}-1} \Delta_{k,l}$
As the regions we removed are adjacent, we have $p = r \pm 1$ and $s = q \pm 1$. But, up to exchanging i and j and/or k and l, without loss of generality $p = r+1$ and $s = q+1$.

Substituting back into the equation, we then conclude: $\Delta_{i,j} = \pm X^{q-r} \Delta_{k,l}$. Therefore the Alexander polynomial is independent of the pair of columns removed.

□

Theorem 2.4.16 (Alexander polynomial): The Alexander polynomial is a knot invariant.

Proof of theorem 2.4.16: It suffices to check that $\Delta_K(X)$ is invariant under the 4 elementary Reidemeister moves.

- Let us first show invariance under $R1a$.
  Let K be a knot with n crossings and D its diagram, in which we index its crossings $c_1, ..., c_n$ and the regions $R_0, ..., R_{n+1}$.
  We consider more particularly two neighbouring regions R_k and R_j that share a strand. We call D' the diagram obtained after carrying out $R1a$ on that strand (see figure 21). As D and D' are identical except on this disc, $A(D')$ is of the form:

$$A(D') = \begin{array}{c} \\ \\ \end{array} \begin{array}{c} \begin{array}{ccc} R_k & R_j & R' \end{array} \\ \begin{pmatrix} A(D)_{R_k,R_j} & \begin{array}{c} * \\ X \end{array} & \begin{array}{c} * \\ -X-1 \end{array} & \begin{array}{c} * \\ 1 \end{array} \\ 0 & & & \end{pmatrix} \end{array}$$

  where $A(D)_{R_k,R_j}$ is the matrix of D from which the columns R_k and R_j have been removed.
  Then, by a block triangular matrix argument, $det(A(D')_{R_k,R_j}) = det(A(D)_{R_k,R_j}) \cdot 1$.



Figure 21: Invariance under $R1a$.

- Invariance under $R1b$ is shown in the same way.
- Let us next show invariance under $R2a$.
  Let us first note a list of operations on a square matrix that preserve the Alexander polynomial of the knot it represents:
  - Transvections (they preserve the determinant).
  - Exchanging two rows or two columns (this multiplies the determinant by $-1 = -X^0$).
  - Multiplying or dividing a row or a column by -1 or by X.

  Up to a renumbering of the regions, we may consider that $R2a$ acts on the regions R_1, R_2 and R_3 and creates the crossings c_1 and c_2:

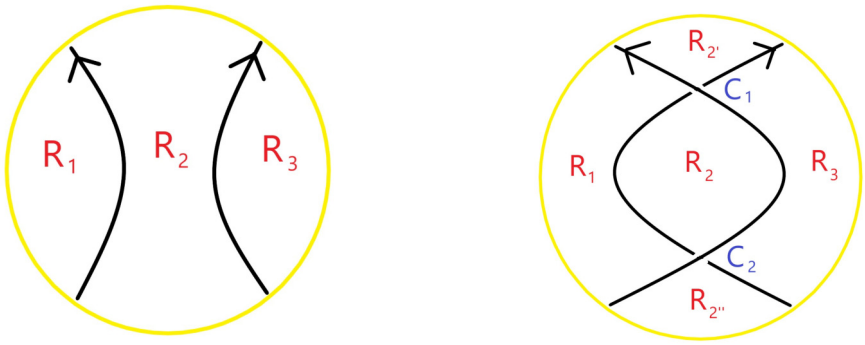


Figure 22: Knots K1 and K2.

After the move, we obtain the matrix

$$A(K2) = \begin{array}{c} \begin{matrix} R_1 & R_2 & R_{2'} & R_{2''} & R_3 & \dots \end{matrix} \\ \begin{pmatrix} -X & 1 & X & 0 & -1 & (0) \\ X & -1 & 0 & -X & 1 & (0) \\ * & (0) & * & * & A(K1)_{1,2} & \end{pmatrix} \end{array}$$

$$\text{Then } A(K2)_{R_1,R_{2'}} = \begin{array}{c} \begin{matrix} R_2 & R_{2''} & R_3 & \dots \end{matrix} \\ \begin{pmatrix} 1 & 0 & -1 & (0) \\ -1 & -X & 1 & (0) \\ (0) & * & A(K1)_{1,2} & \end{pmatrix} \end{array}$$

$$\text{We have } \Delta_{K2} = det \begin{pmatrix} 1 & 0 & -1 & (0) \\ 0 & -X & 0 & (0) \\ (0) & * & A(K1)_{1,2} & \end{pmatrix} \quad (L_2 \leftarrow L_2 + L_1)$$

$$\Delta_{K2} = det \begin{pmatrix} -X & 0 & (0) \\ * & A(K1)_{1,2} & \end{pmatrix} \quad \text{(expansion along } C_1)$$

$$\Delta_{K2} = det \begin{pmatrix} 1 & 0 & (0) \\ * & A(K1)_{1,2} & \end{pmatrix} \quad (L_1 \leftarrow \frac{L_1}{-X})$$

$$\Delta_{K2} = det \begin{pmatrix} A(K1)_{1,2} \end{pmatrix} \quad \text{(expansion along } L_1)$$

Therefore $\Delta_{K2} = \Delta_{K1}$

- Finally, let us show invariance under $R3a$. To do so, we shall need a lemma:
Let K be a knot and A its matrix. Then $\Delta(A) = \Delta(A_+)$ where A_+ is the matrix with general term $(A_+)_{i,j} = |a_{i,j}|$. (If $A_{i,j}$ is the polynomial $\sum a_k X^k$, then $(A_+)_{i,j}$ will be the polynomial $\sum |a_k| X^k$.)
Indeed, let us recall the "number" $n(R)$ of a region. One may notice that at a crossing, one can associate the parity of $n(R)$ with the sign of the coefficient of the region R at that crossing. Consider the following crossing:

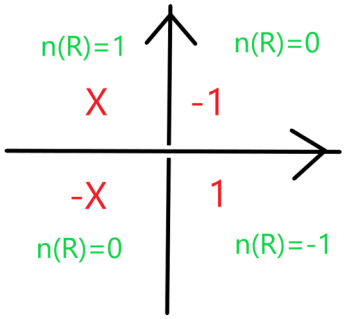


Figure 23: Coefficients and "numbers" around a crossing.

We see that the regions with a negative coefficient have an even $n(R)$ and that the regions with a positive coefficient have an odd $n(R)$.

We may then multiply by -1 the columns of A corresponding to the regions having an odd "number", which gives us a matrix A' satisfying $\Delta(A') = \Delta(A)$.
Moreover, all the coefficients around a crossing now have the same sign, so it suffices to multiply all the rows with negative coefficients by -1 in order to obtain the desired result.
Of course, we likewise have $\Delta(B) = \Delta(B_+)$, where B is obtained by deleting 2 columns of adjacent regions from A.

We can now prove invariance under $R3a$. Let us take figures similar to those in the proof of invariance under $R2a$:



Figure 24: Knots K1 and K2.

Before the move, we therefore have the matrix

$$A(K1) = \begin{array}{c} \begin{matrix} R_1 & R_2 & R_3 & R_4 & R_5 & R_6 & R_7 & \dots \end{matrix} \\ \begin{pmatrix} X & -X & 1 & -1 & 0 & 0 & 0 & (0) \\ 0 & X & -1 & 0 & -X & 1 & 0 & (0) \\ 0 & 0 & 1 & -X & 0 & -1 & X & (0) \\ M_1 & & (0) & & & M_2 & & \end{pmatrix} \end{array}$$

After the move, we have the matrix

$$A(K2) = \begin{array}{c} \begin{matrix} R_{1'} & R_{2'} & R_{3'} & R_{4'} & R_{5'} & R_{6'} & R_{7'} & ... \end{matrix} \\ \begin{pmatrix} 0 & 0 & X & 0 & -X & 1 & -1 & (0) \\ -X & 1 & X & 0 & -1 & 0 & 0 & (0) \\ X & 0 & -X & -1 & 0 & 0 & 1 & (0) \\ M_1 & & (0) & & & M_2 & & \end{pmatrix} \end{array}$$

(M_1 and M_2 are submatrices common to both matrices.)

Let us show that $\Delta(A(K1)) = \Delta(A(K2))$. To do so, we shall delete the columns 4 and 7 of both matrices (they correspond to adjacent regions) and show that $\Delta(A(K1)_{R_4,R_7}) = \Delta(A(K1)_{R_{4'},R_{7'}})$. Set $B1 = \Delta(A(K1)_{R_4,R_7})$ and $B2 = \Delta(A(K1)_{R_{4'},R_{7'}})$ and let us show that $\Delta(B1_+) = \Delta(B2_+)$, which will suffice to conclude by the lemma.

We have $B1_+ = \begin{pmatrix} X & X & 1 & 0 & 0 & (0) \\ 0 & X & 1 & X & 1 & (0) \\ 0 & 0 & 1 & 0 & 1 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix}$ and $B2_+ = \begin{pmatrix} 0 & 0 & X & X & 1 & (0) \\ X & 1 & X & 1 & 0 & (0) \\ X & 0 & X & 0 & 0 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix}$

$$\Delta(B2_+) = det \begin{pmatrix} -X & 0 & X & X & 1 & (0) \\ 0 & 1 & X & 1 & 0 & (0) \\ 0 & 0 & X & 0 & 0 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (C_1 \leftarrow C_1 - C_3)$$

$$\Delta(B2_+) = det \begin{pmatrix} -X & 0 & 1 & X & 1 & (0) \\ 0 & 1 & 1 & 1 & 0 & (0) \\ 0 & 0 & 1 & 0 & 0 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (C_3 \leftarrow \frac{C_3}{X})$$

$$\Delta(B2_+) = det \begin{pmatrix} -X & 0 & 1 & X & 1 & (0) \\ 0 & X & X & X & 0 & (0) \\ 0 & 0 & 1 & 0 & 0 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (L_2 \leftarrow X \times L_2)$$

$$\Delta(B2_+) = det \begin{pmatrix} -X & 0 & 0 & X & 1 & (0) \\ 0 & X & X & X & 0 & (0) \\ 0 & 0 & 1 & 0 & 0 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (L_1 \leftarrow L_1 - L_3)$$

$$\Delta(B2_+) = det \begin{pmatrix} -X & -X & -X & 0 & 1 & (0) \\ 0 & X & X & X & 0 & (0) \\ 0 & 0 & 1 & 0 & 0 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (L_1 \leftarrow L_1 - L_2)$$

$$\Delta(B2_+) = det \begin{pmatrix} -X & -X & -X & 0 & 1 & (0) \\ 0 & X & X & X & 0 & (0) \\ 0 & 0 & X & 0 & 0 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (L_3 \leftarrow X \times L_3)$$

$$\Delta(B2_+) = det \begin{pmatrix} -X & -X & -X & 0 & 1 & (0) \\ 0 & X & X & X & 0 & (0) \\ -X & -X & 0 & 0 & 1 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (L_3 \leftarrow L_3 + L_1)$$

$$\Delta(B2_+) = det \begin{pmatrix} -X & -X & -1 & 0 & 1 & (0) \\ 0 & X & 1 & X & 0 & (0) \\ -X & -X & 0 & 0 & 1 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (C_3 \leftarrow \frac{C_3}{X})$$

$$\Delta(B2_+) = det \begin{pmatrix} X & X & 1 & 0 & -1 & (0) \\ 0 & X & 1 & X & 0 & (0) \\ -X & -X & 0 & 0 & 1 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (L_1 \leftarrow -L_1)$$

$$\Delta(B2_+) = det \begin{pmatrix} X & X & 1 & 0 & -1 & (0) \\ 0 & X & 1 & X & 0 & (0) \\ 0 & 0 & 1 & 0 & 0 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (L_3 \leftarrow L_3 + L_1)$$

$$\Delta(B2_+) = det \begin{pmatrix} X & X & 1 & 0 & 0 & (0) \\ 0 & X & 1 & X & 1 & (0) \\ 0 & 0 & 1 & 0 & 1 & (0) \\ M'1_+ & & (0) & & M'2_+ & \end{pmatrix} (C_5 \leftarrow C_5 + C_3)$$

Whence $\Delta(B2_+) = \Delta(B1_+)$, which completes the proof: the Alexander polynomial is indeed a knot invariant!

□

Unfortunately, the Alexander polynomial is not a complete knot invariant. For example, it does not distinguish the right trefoil knot from the left trefoil knot.

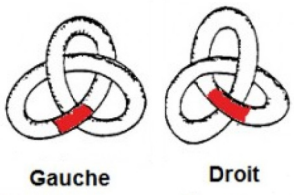


Figure 25: Left trefoil knot T_1 and right trefoil knot T_2.

2.4.3 SKEIN RELATIONS AND FINER INVARIANTS

The idea here is to present a polynomial approach alternative to Alexander's, more focused on links. Thanks to what we shall call *skein* relations, it will be possible to compute more efficiently invariants more advanced than Alexander's.

Definition 2.4.17: A Conway triple is defined as a triple (L^+, L^-, L^0) of oriented links of $\mathbb{R}^3$ that have identical diagrams outside a disc, where they differ as in figure 26:

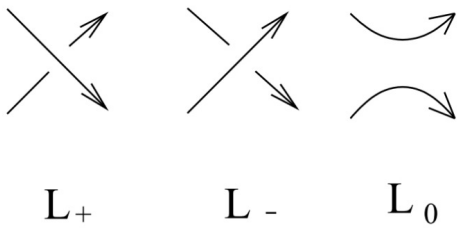


Figure 26: Conway triple.

Definition 2.4.18 (*Skein* relation): We say that a link invariant F (with values in a ring $\mathbb{A}$) satisfies a *skein* relation when there exists $(a, b, c) \in \mathbb{R}^3$ such that for every Conway triple (L_+, L_-, L_0), we have:

$$aF(L_+) + bF(L_-) = cF(L_0)$$

Remark 2.4.19: When one wants to compute the invariant of a link L through a *skein* relation, one may consider one of its resolving trees. It consists of a binary tree built in the following way:

- The root is the link L.
- For each node of the tree, its children are built as follows:
  - We call L' the link corresponding to the node of the tree.
  - We choose a crossing of L' and identify it: L_+ or L_-.
  - The children of the node will be the links associated with the other members of the Conway triple.
- The leaves of the tree are trivial links (with no crossing).

We justify that such a construction "algorithm" terminates thanks to the following property.

Proposition 2.4.20 (stated without proof): For every link diagram D, there exist crossings of D such that, on inverting them, one obtains a diagram of $A \sqcup ... \sqcup A$.
Although admitted, this proposition is very intuitive.

The result follows immediately. Indeed, write $\mathbf{C} = c_{i_1}, ..., c_{i_p}$ for the set of crossings of D such that, on inverting them, one obtains a diagram of trivial links, and $n = |\mathbf{C}|$.
Suppose that $n = 0$. Then $D = A \sqcup ... \sqcup A$: the algorithm has terminated.
Now let $j \in [\![1, p]\!]$, and consider c_{i_j}. It is either L_+ or L_-. We assume here without loss of generality that it is L_+.
By inverting L_+ into L_-, we decrease the number of crossings to be inverted by 1 (because c_{i_j} no longer has to be done): $n \leftarrow n - 1$. As for the second branch of the tree, the one giving L_0, we note that L_0 is not a crossing. Here too, we have one fewer crossing to invert ($n \leftarrow n - 1$). The cardinality of $\mathbf{C}$ being a strictly decreasing function with values in $\mathbb{N}$, the algorithm will terminate.

Example 2.4.21: Resolving tree of the figure-eight knot H:

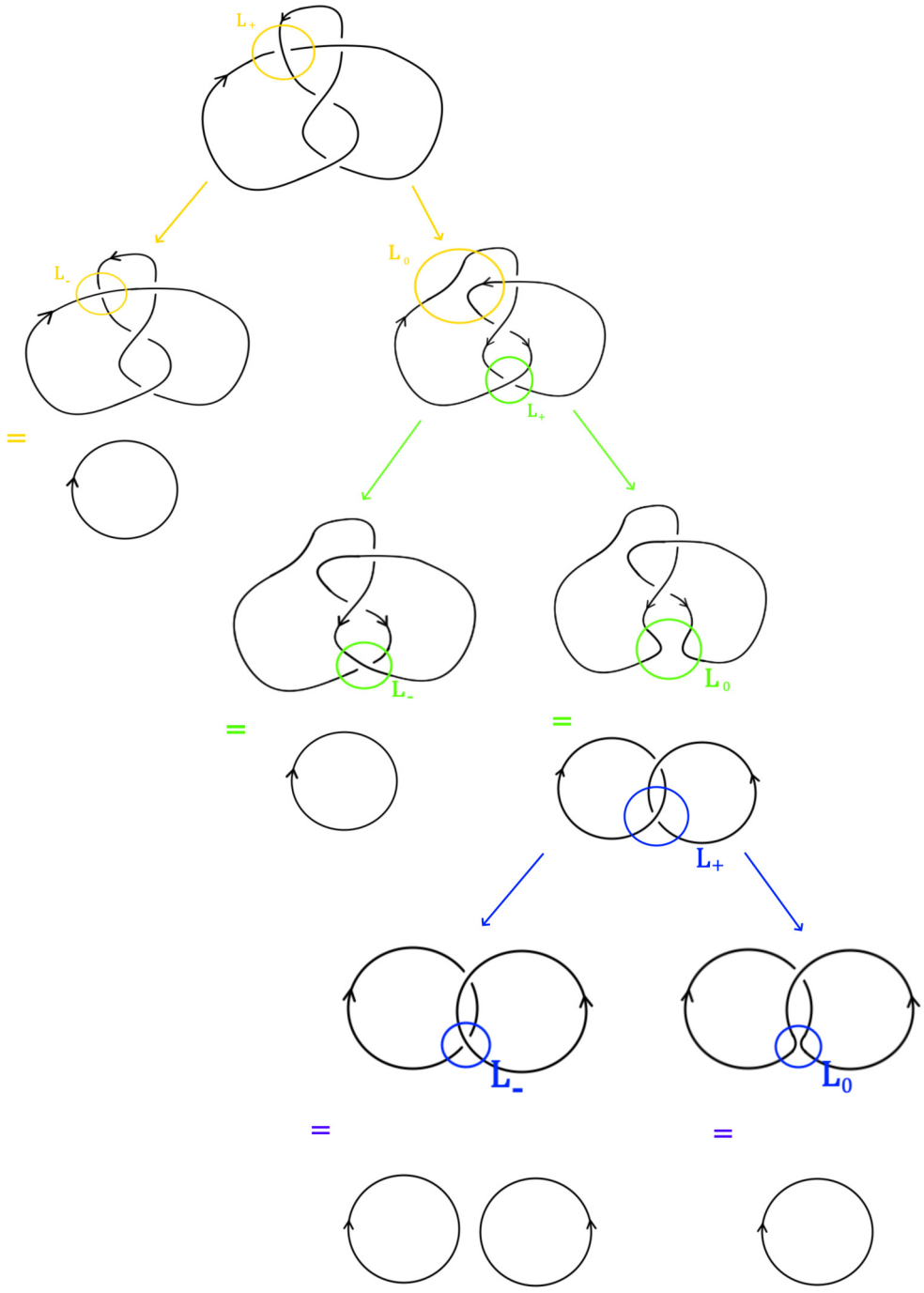


Figure 27: Example of a resolving tree for the figure-eight knot.

We can now define polynomial invariants using *skein* relations. These definitions have the advantage of being far easier to handle than the original definitions of their authors.

For example, here is the definition of the Alexander polynomial that is the most used in practice:

Definition 2.4.22: The Alexander polynomial of links $L \mapsto \Delta_L(X) \in \mathbb{Z}[\sqrt{X}, \frac{1}{\sqrt{X}}]$ is the Laurent polynomial in $\sqrt{X}$ satisfying the following 3 properties:

- $\Delta_.$ is a link invariant.
- $\Delta_A = 1$
- For every Conway triple (L_+, L_-, L_0), we have:

$$\Delta_{L_+}(X) - \Delta_{L_-}(X) = (\sqrt{X} - \frac{1}{\sqrt{X}}) \cdot \Delta_{L_0}(X)$$

We *accept without proof* that such a definition does coincide with the matrix definition.

Remark 2.4.23: This new definition has both advantages and drawbacks. Some advantages are:

- The computation is carried out entirely from the diagrams.
- It is convenient for establishing recurrence formulas on infinite links (or families of links).

It has, however, certain drawbacks:

- The number of leaves of the resolving tree may be *exponential* in the number of crossings of the diagram of the initial knot, whereas the matrix computation was *polynomial* (determinant).
- A certain skill is needed in order to choose the crossings that lead to the most efficient possible resolving tree.
- Sometimes it can be difficult to identify the trivial knots and links that one arrives at.

Proposition 2.4.24: Let K, J be two knots. The Alexander polynomial satisfies:

i) $\Delta_{J\#K}(X) = \Delta_J(X) \cdot \Delta_K(\mathrm{X})$

ii) $\Delta_{K \sqcup A} = 0$

Proof of proposition 2.4.24:
i) has already been proved.
Let K be a knot. The boundary between K and A may be seen as a crossing of type L_0. One can then deduce from it the knots corresponding to the crossings L_+ and L_-.
We then note that these last two knots are isotopic to K through $R1$. Applying the *skein* relation of the Alexander polynomial, it follows that:

$$\Delta_K(X) - \Delta_K(X) = (\sqrt{X} - \frac{1}{\sqrt{X}}) \cdot \Delta_{K \sqcup A}(X)$$

Then,

$$\Delta_{K \sqcup A}(X) = 0$$

□

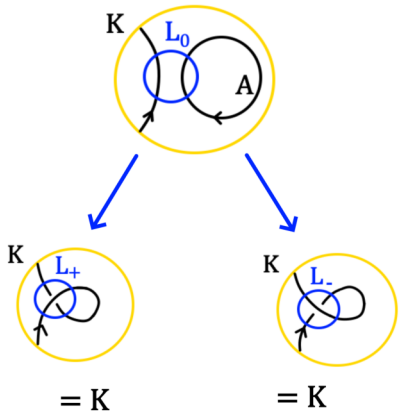


Figure 28: Illustration of the proof of proposition 2.4.24.

Remark 2.4.25: Property ii) of the Alexander polynomial generalises to links.
Indeed, let L be a link. Let us write L in the form $K_1 \sqcup ... \sqcup K_n$.
We then apply the same reasoning as in the previous proof to some K_i with $i \in [\![1, n]\!]$.
On the other hand, we have not defined a "connected sum" on links: it is therefore impossible to extend the result i).

Definition 2.4.26 (Conway polynomial): The Conway polynomial of links $L \mapsto \nabla_L(X) \in \mathbb{Z}[X]$ is the polynomial in X satisfying the following 3 properties:

- $\nabla_.$ is a link invariant.
- $\nabla_A = 1$
- For every Conway triple (L_+, L_-, L_0), we have:

$$\nabla_{L_+}(X) - \nabla_{L_-}(X) = X \cdot \nabla_{L_0}(X)$$

Justification of definition 2.4.26: Let us check that the Conway polynomial thus defined is indeed a link invariant. We could prove invariance under $R1a$, $R1b$, $R2a$ and $R3a$, but it is possible to reduce the work of the proof drastically thanks to the following clever property.

Remark 2.4.27: We have $\Delta_.(X) = \nabla_.(\sqrt{X} - \frac{1}{\sqrt{X}})$

Justification of remark 2.4.27: To show that these two polynomials are equal, we must check that they do coincide on every link. Now they are entirely characterised by their image of the unknot and by their *skein* relation. It then suffices to check that they coincide on these two points.
We do have $\Delta_A = \nabla_A = 1$. Let (L_+, L_-, L_0) be a Conway triple. Set $Y = \sqrt{X} - \frac{1}{\sqrt{X}}$. We have:

$$\nabla_{L_+}(Y) - \nabla_{L_-}(Y) = Y \cdot \nabla_{L_0}(Y)$$
$$\textit{i.e.} \quad \nabla_{L_+}(Y) - \nabla_{L_-}(Y) = (\sqrt{X} - \frac{1}{\sqrt{X}}) \cdot \nabla_{L_0}(Y)$$

Therefore $\nabla_.(Y)$ and $\Delta_.(X)$ share the same *skein* relation.
Finally, they are equal.

The invariance of the Conway polynomial therefore follows directly from that of the Alexander polynomial.

□

Proposition 2.4.28: Let K, J be two knots. The Conway polynomial satisfies:

i) $\nabla_{J\#K}(X) = \nabla_J(X) \cdot \nabla_K(X)$

ii) $\nabla_{K\sqcup A} = 0$

Proof of proposition 2.4.28: The results follow immediately from remark 2.4.27 and from the properties of the Alexander polynomial stated in proposition 2.4.24. (For i), it suffices to write $\nabla_{J\#K}(\sqrt{X} - \frac{1}{\sqrt{X}})$.)

□

Remark 2.4.29: Here too, one can generalise ii) to the disjoint union of a link and A.

Example 2.4.29: The Conway polynomial of the figure-eight knot.
We write H for the figure-eight knot. We shall here use the resolving tree proposed in figure 27 to compute $\nabla_H(X)$. We number the various diagrams obtained from top to bottom and from left to right: from D_1 to D_7. We first consider the yellow triple:

$$\begin{aligned}\nabla_{D_1}(X) - \nabla_{D_2}(X) = X\nabla_{D_3}(X) \Leftrightarrow \nabla_{D_1}(X) &= \nabla_{D_2}(X) + X\nabla_{D_3}(X)\\ &= \nabla_A(X) + X\nabla_{D_3}(X)\\ &= 1 + X\nabla_{D_3}(X)\end{aligned}$$

Then the green triple:

$$\begin{aligned}\nabla_{D_3}(X) - \nabla_{D_4}(X) = X\nabla_{D_5}(X) \Leftrightarrow \nabla_{D_3}(X) &= \nabla_{D_4}(X) + X\nabla_{D_5}(X)\\ &= \nabla_A(X) + X\nabla_{D_5}(X)\\ &= 1 + X\nabla_{D_5}(X)\end{aligned}$$

And finally the blue triple:

$$\begin{aligned}\nabla_{D_5}(X) - \nabla_{D_6}(X) = X\nabla_{D_7}(X) \Leftrightarrow \nabla_{D_5}(X) &= \nabla_{D_6}(X) + X\nabla_{D_7}(X)\\ &= \nabla_{D_{A\sqcup A}}(X) + X\nabla_A(X)\\ &= X\end{aligned}$$

Finally, it follows that:

$$\begin{aligned}\nabla_H(X) = \nabla_{D_1}(X) &= 1 + X(1 + X(X))\\ &= 1 + X + X^3\end{aligned}$$

We now propose to introduce a major polynomial introduced by Vaughan Jones, which revived the study of knot theory at a time when it had lost its momentum: the Jones polynomial. Its importance, however, goes far beyond that framework: it has applications in many areas of mathematics and physics, such as operator models, statistical groups and quantum models.
In this report we shall be interested in 2 definitions (neither of which is the one originally envisaged by the author of the polynomial, the latter being far too complex to be treated in depth here), which may be seen as complementary: first an approach introduced by Louis Kauffman that may be compared to the

statistical model, then a definition similar to that of the Conway polynomial, through a *skein* relation.
Kauffman's idea is fairly visual and intuitive at first, which is why we shall content ourselves initially with *explaining* the manipulations to be undertaken, without seeking a rigorous formulation of the Jones polynomial, which will in any case lie in its second definition.
His idea consists in representing the various possible "states" of an unoriented knot. Indeed, consider an arbitrary unoriented knot K. Each crossing then locally divides the plane into two "complementary" angles: an angle of type A which is located to the right of the over-strand of a crossing, and an angle of type B which is located to the right of the under-strand (see figure 29).

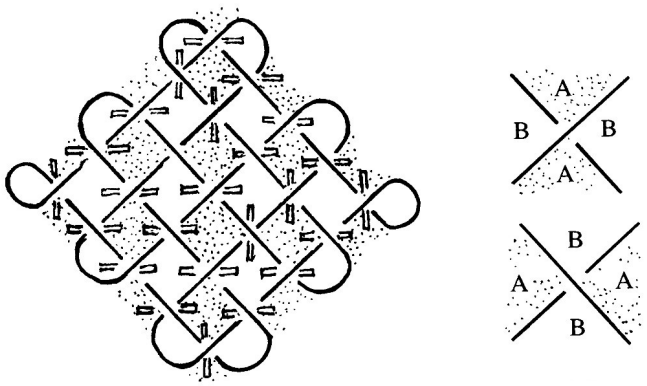


Figure 29: State of a knot (angles of types A and B).

It is then possible to associate with each crossing what we shall call a *Kauffman spin*: it may be of type *up* or of type *down*. A state of the knot then represents the characterisation of all the crossings by a *spin*. For example, in figure 29, a *spin* is represented by a stick: if it is in the zone of angle A, it is an *up*, otherwise it is a *down*.
We then introduce the notion of resolution of a state of K. This operation has a strong analogy with the transformation of an L_+ or L_- into an L_0 (for a given Conway triple). Indeed, given a state of K, we shall rearrange the strands of the knot in accordance with the *spin*: if it is of type *up*, we join the two zones of angle of type A so as to form only one, otherwise we join those of type B (see figure 30).

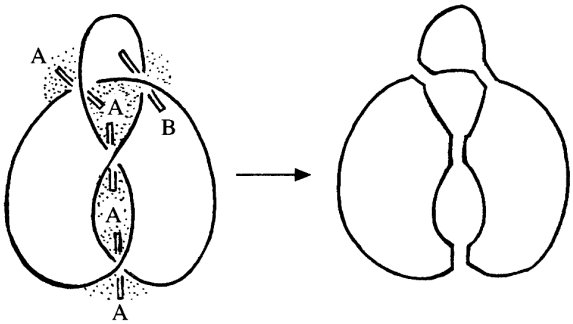


Figure 30: The resolution of a state of the figure-eight knot.

These definitions generalise naturally to links: since angles of type A or B are only defined in a neighbourhood of the crossing under consideration, there will be no conflict over the various regions of $\mathbb{R}^3$ delimited by the link. It is then possible to define the Kauffman bracket:

Definition 2.4.30 (Kauffman bracket): Let L be an unoriented link with n crossings. We write $S(L)$ for the finite set of states of L, of cardinality 2^n. We then define the Kauffman bracket as a Laurent polynomial in a:

$$\langle L \rangle = \sum_{s \in S(L)} a^{\alpha(s)-\beta(s)}(-a^2 - a^{-2})^{\gamma(s)-1}$$

where, for $s \in S(L)$, $\alpha(s)$ denotes the number of crossings of type *up*, $\beta(s)$ the number of crossings of type *down*, and $\gamma(s)$ the number of closed curves obtained when the state s is resolved.

Definition 2.4.31: A quadruple of links is called a Kauffman quadruple when they are identical at every point except within a disc, as shown in figure 31.
We then write them respectively $(D_+, D_-, D_0, D_\infty)$:

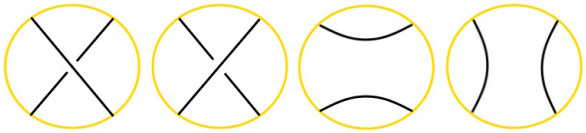


Figure 31: Kauffman quadruple.

In what follows, we shall speak of a Kauffman triple when we choose to consider only 3 of the 4 possible forms of the quadruple. These will then be made explicit.

Proposition 2.4.32: The Kauffman bracket satisfies the following relations:

i) $\langle A \rangle = 1$

ii) $\langle L \sqcup A \rangle = (-a^2 - a^{-2})\langle L \rangle$

iii) Let (D_+, D_0, D_∞) be a Kauffman triple. Then:

$$\langle D_+ \rangle = a\langle D_0 \rangle + a^{-1}\langle D_\infty \rangle$$

iv) Let (D_-, D_∞, D_0) be a Kauffman triple. Then:

$$\langle D_- \rangle = a\langle D_\infty \rangle + a^{-1}\langle D_0 \rangle$$

Proof of proposition 2.4.32: i) is immediate.
To show ii), it suffices to realise that $\langle L \rangle$ and $\langle L \sqcup A \rangle$ are identical except that for each state s, $L \sqcup A$ has one more curve: $\gamma_L(s) = \gamma_{L \sqcup A}(s) + 1$. We can therefore factor the sum of $\langle L \sqcup A \rangle$ by $(-a^2 - a^{-2})$, after which the result is immediate.
Next, let us show iii). Let (D_+, D_0, D_∞) be a Kauffman triple. We call c_0 the crossing of D_+ that does not exist in D_0 and D_∞.
Let $s \in S(D_+)$. Then c_0 can be resolved according to 2 configurations only: those that lead (inside a disc in a neighbourhood of c_0) to D_0 and to D_∞. It follows that $S(D_+) = S(D_0) \sqcup S(D_\infty)$.
We note that whatever the configuration of c_0, the final number of curves $\gamma(s)$ is the same for the 3 knots. We therefore have

$$\langle D_+ \rangle = \sum_{s \in S(D_0)} a^{\alpha(s)-\beta(s)}(-a^2 - a^{-2})^{\gamma(s)-1} + \sum_{s \in S(D_\infty)} a^{\alpha(s)-\beta(s)}(-a^2 - a^{-2})^{\gamma(s)-1}$$

Then for each state $s \in S(D_0)$, the number of *spin up* of D_+ is $\alpha_{D_+}(s) = \alpha_{D_0}(s) + 1$. On the other hand, $\beta_{D_+}(s) = \beta_{D_0}(s)$. Therefore

$$\begin{aligned}\forall s \in S(D_0), \sum_{s\in S(D_0)} a^{\alpha_{D_+}(s)-\beta_{D_+}(s)}(-a^2-a^{-2})^{\gamma(s)-1} &= \sum_{s\in S(D_0)} a \cdot a^{\alpha_{D_0}(s)-\beta_{D_0}(s)}(-a^2-a^{-2})^{\gamma(s)-1} \\ &= a\langle D_0\rangle\end{aligned}$$

Symmetrically,

$$\begin{aligned}\forall s \in S(D_\infty), \sum_{s\in S(D_\infty)} a^{\alpha_{D_+}(s)-\beta_{D_+}(s)}(-a^2-a^{-2})^{\gamma(s)-1} &= \sum_{s\in S(D_\infty)} a^{-1} \cdot a^{\alpha_{D_\infty}(s)-\beta_{D_\infty}(s)}(-a^2-a^{-2})^{\gamma(s)-1} \\ &= a^{-1}\langle D_\infty\rangle\end{aligned}$$

Finally, we do have $\langle D_+\rangle = a\langle D_0\rangle + a^{-1}\langle D_\infty\rangle$
Lastly, to show iv), it suffices to notice that D_- can be thought of as D_+ up to a rotation (on a local disc, therefore). Then, one can express the bracket of the Kauffman triple of (D_+, D_0, D_∞). Next, it is a matter of "untwisting" the 3 links in a neighbourhood of the disc under consideration. We keep the same relation, but with (D_-, D_∞, D_0).
From this last point we shall draw a very important consequence: when we consider a crossing of an unoriented link, it is impossible to know whether it is a D_+ or a D_-. However, once that choice has been made, the rest of the triple and therefore of the Kauffman bracket is determined *uniquely*.

□

These 4 properties satisfied by the Kauffman bracket resemble the definitions of the Alexander and Conway polynomials (in particular the pseudo *skein* relation): could it also be a link invariant?
Unfortunately, the following example proves that it is not:

Example 2.4.33: The Kauffman bracket of the knot with one crossing.

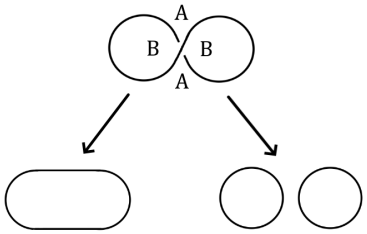


Figure 32: Resolved states of the knot with one crossing.

We deduce from figure 32 that:

$$\begin{aligned}\langle K\rangle &= a^{0-1}(-a^2-a^{-2})^0 + a^{1-0}(-a^2-a^{-2})^1 \\ &= a^{-1} - a^3 - a^{-1} \\ &= -a^3\end{aligned}$$

Thus, although the knot with one crossing is isotopic to the unknot, they do not have the same Kauffman bracket: the latter *is not* a link invariant.
Here are the properties of the Kauffman bracket that lie behind this example:

Theorem 2.4.34:

a) The Kauffman bracket is invariant under $R2$ and $R3$.

b) The move $R1$ multiplies the bracket by $-a^{\pm 3}$.
More precisely:

- Applying $R1a$ or $R1b$ multiplies it by $-a^3$.
- Applying $R1c$ or $R1d$ multiplies it by $-a^{-3}$.

Proof of theorem 2.4.34: Throughout the proof, we consider a link L.
Let us first show that $\langle \cdot \rangle$ is invariant under $R2$.
We write L_{R2} for the link obtained after applying $R2$ to L.
Relying on figure 33, we number the various links obtained from top to bottom and from right to left: from L_1 to L_7. Using the various relations of the Kauffman bracket, we can then write:

$$\begin{aligned}
\langle L_{R2} \rangle = \langle L_1 \rangle &= a\langle L_2 \rangle + a^{-1}\langle L_3 \rangle \\
&= a(a\langle L_4 \rangle + a^{-1}\langle L_5 \rangle) + a^{-1}(a\langle L_6 \rangle + a^{-1}\langle L_7 \rangle) \\
&= a^2\langle L_4 \rangle + \langle L_4 \sqcup A \rangle + \langle L_6 \rangle + a^{-2}\langle L_4 \rangle \\
&= [a^2 + (-a^2 - a^{-2}) + a^{-2}]\langle L_4 \rangle + \langle L_6 \rangle \\
&= \langle L_6 \rangle = \langle L \rangle
\end{aligned}$$

$\langle \cdot \rangle$ is indeed invariant under $R2$.

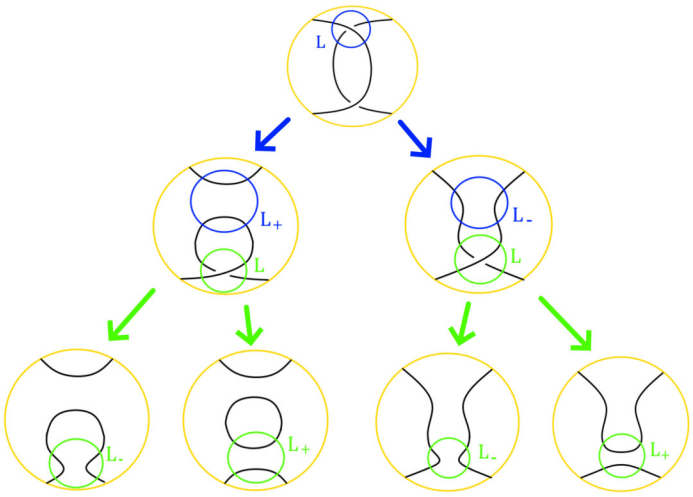


Figure 33: Illustration of the invariance of $\langle \cdot \rangle$ under $R2$.

Let us now show invariance under $R3$.
This time we rely on figure 34, whose diagrams we number according to the same convention as before but in batches of three: from L_1 to L_6.

It follows that:

$$\begin{cases} \langle L_1 \rangle = a\langle L_2 \rangle + a^{-1}\langle L_3 \rangle \\ \langle L_4 \rangle = a\langle L_5 \rangle + a^{-1}\langle L_6 \rangle \end{cases}$$

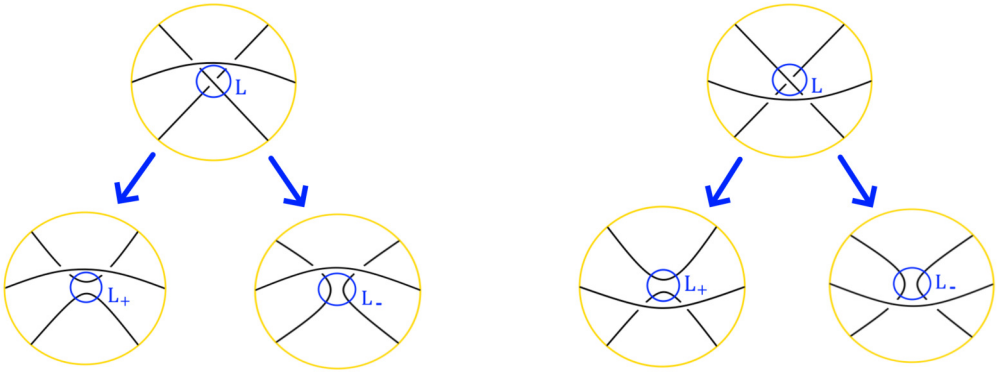


Figure 34: Illustration of the invariance of $\langle \cdot \rangle$ under $R3$.

Now figures 35 and 36 show precisely that, on the one hand, $\langle L_2 \rangle = \langle L_5 \rangle$ through the invariance of $\langle \cdot \rangle$ under $R2$ that we have just proved, and, on the other hand, that $\langle L_3 \rangle = \langle L_6 \rangle$ because their diagrams are isotopic in the plane.
Finally, $\langle L_1 \rangle$ and $\langle L_4 \rangle$ are equal term by term: they are therefore equal to each other.
$\langle \cdot \rangle$ is indeed invariant under $R3$.

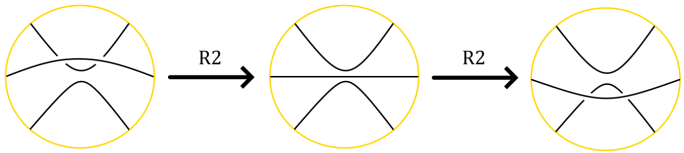


Figure 35: Equivalence of the brackets under R2.

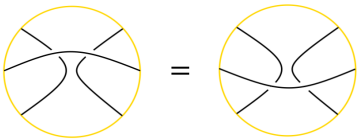


Figure 36: Isotopic links.

Let us finally show that $R1$ multiplies $\langle \cdot \rangle$ by $-a^{\pm 3}$.
Let us show that $R1a$ multiplies $\langle \cdot \rangle$ by $-a^3$. The arguments for $R1b$, $R1c$ and $R1d$ are symmetric.
We rely on figure 37, following the same numbering convention as before: from L_1 to L_3, where $L_1 = L'$

is the image of $L = L_3$ under $R1$. It follows that:

$$\begin{aligned} \langle L_1 \rangle &= a\langle L_2 \rangle + a^{-1}\langle L_3 \rangle \\ i.e.\ \ \langle L' \rangle &= [a(-a^2 - a^{-2}) + a^{-1}]\langle L \rangle \\ &= -a^3\langle L \rangle \end{aligned}$$

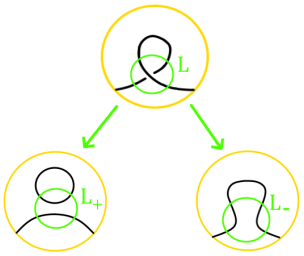


Figure 37: Kauffman triple around $R1$.

□

In order to arrive at a link invariant, we must therefore find a way of making this coefficient $-a^{\pm 3}$, a residue of the first Reidemeister move, "disappear". We then introduce:

Definition 2.4.35 (The *writhe*): The *writhe* of a link L, written $w(L)$, is defined as the sum of the signs of the crossings of L: writing $n(c_-)$ for the number of negative crossings and $n(c_+)$ for the number of positive crossings, $w(L) = n(c_+) - n(c_-)$.

Theorem 2.4.36: Recall that we have written $\mathfrak{L}$ for the set of links. Then:

$$\begin{array}{ccccc} V & : & \mathfrak{L} & \longrightarrow & \mathbb{Z}[X, X^{-1}] \\ & & L & \longmapsto & V_L(a) = (-a)^{-3w(L)} \cdot \langle |L| \rangle \end{array}$$

(where $\langle |L| \rangle$ denotes the Kauffman bracket of the unoriented link L) is a link invariant: this is (almost) the Jones polynomial!

Proof of theorem 2.4.36: We already have that $\langle \cdot \rangle$ is invariant under $R2$ and $R3$, so *a fortiori* it is invariant under $R2a$ and $R3a$. Let us next show that the *writhe* is also invariant under these moves, which will allow us to conclude for $R2a$ and $R3a$. For greater clarity, we recall in figure 38 the 4 elementary Reidemeister moves:

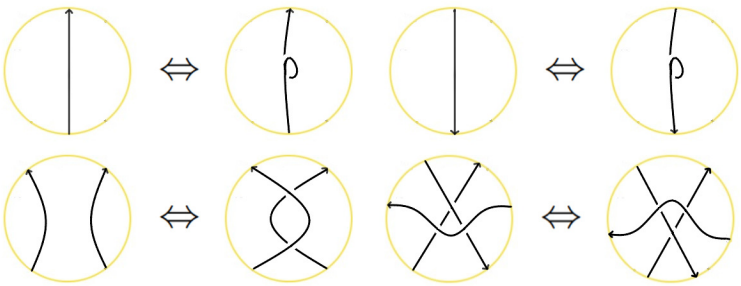


Figure 38: The 4 elementary Reidemeister moves: $R1a$, $R1b$, $R2a$ and $R3a$.

We then note that the *writhe* is invariant under $R2a$ and $R3a$. Indeed, for each positive crossing created, a negative crossing follows ($R2a$), or else the crossings simply exchange positions ($R3a$).
Let us finally show that $V_.(a)$ is invariant under $R1a$.
We note that for every link L, the image of its composition by $R1a$, which we write L', has one additional *positive* crossing: $w(L') = w(L) + 1$.
Then it follows that:

$$\begin{aligned} V_{L'}(a) &= (-a)^{-3w(L')} \cdot \langle |L'| \rangle \\ &= [(-a)^{-3}(-a)^{-3w(L)}] \cdot \langle |L'| \rangle \\ &= (-a)^{-3}(-a)^3 \cdot V_L(a) \\ &= V_L(a) \end{aligned}$$

$V_.(a)$ is indeed invariant under $R1a$. The argument proving invariance under $R1b$ is rigorously symmetric: there too $w(L') = w(L) + 1$ and $\langle |L'| \rangle = -a^{-3}\langle |L| \rangle$.

□

Remark 2.4.37 (The "real" Jones polynomial): In order to obtain the authentic Jones polynomial from $V_.(a)$, it suffices to carry out a change of notation: $J_.(X) = V_.(a^{-4})$.

Kauffman's idea therefore leads to the Jones polynomial. However, although the Kauffman bracket has a pseudo *skein* relation, determining the *writhe* and hence $V_.(a)$ can quickly turn out to be tedious. We propose here to give a definition using a *skein* relation that is easier to use in practice.

Definition 2.4.38 (Jones polynomial): The Jones polynomial of links $L \mapsto J_L(X) \in \mathbb{Z}[\sqrt{X}, \frac{1}{\sqrt{X}}]$ is the Laurent polynomial in $\sqrt{X}$ satisfying the following 3 properties:

- $J_.$ is a link invariant.
- $J_A(X) = 1$
- For every Conway triple (L_+, L_-, L_0), we have:

$$\frac{1}{X} \cdot J_{L_+}(X) - X \cdot J_{L_-}(X) = (\sqrt{X} - \frac{1}{\sqrt{X}}) \cdot J_{L_0}(X)$$

Justification of definition 2.4.38: Let us check that this new definition does coincide with the definition using the Kauffman bracket.
It therefore suffices to show that for every link L, $V_L(a^4)$ satisfies the same *skein* relation as $J_L(X^{-1})$ (since we of course have that $V_.(a^4)$ is a link invariant and that $V_A(a^4) = 1$), which becomes: $X \cdot J_{L_+}(X^{-1}) - X^{-1} \cdot J_{L_-}(X^{-1}) = (\frac{1}{\sqrt{X}} - \sqrt{X}) \cdot J_{L_0}(X^{-1})$.
Let (L_+, L_-, L_0) be a Conway triple. We then consider $(D_+, D_-, D_0, D_\infty)$ the associated Kauffman quadruple such that $(D_+, D_-) = (|L_+|, |L_-|)$. From proposition 2.4.32 we draw:

$$\begin{cases} \langle D_+ \rangle = a\langle D_0 \rangle + a^{-1}\langle D_\infty \rangle \\ \langle D_- \rangle = a\langle D_\infty \rangle + a^{-1}\langle D_0 \rangle \end{cases}$$

By multiplying these equations by a and a^{-1} respectively, then subtracting them, we obtain:

$$a\langle D_+\rangle - a^{-1}\langle D_-\rangle = (a^2 - a^{-2})\langle D_0\rangle$$

Now $V_{L_+}(a) = (-a)^{-3w(L_+)} \cdot \langle |L_+| \rangle$
Therefore $\langle D_+\rangle = (-a)^{3w(L_+)} \cdot V_{L_+}(a)$. Similarly, $\langle D_-\rangle = (-a)^{3w(L_-)} \cdot V_{L_-}(a)$.
Now the *writhes* of L_+ and L_- differ by a factor 2. Indeed, $w(L_+) - 1 = w(L_-) + 1 = w(L_0) = w$.
Finally, simplifying the last equation presented by $(-a)^{3w}$, we have:

$$\begin{aligned} a(-a)^3 V_{L_+}(a) - a^{-1}(-a)^{-3} V_{L_-}(a) &= (a^2 - a^{-2}) V_{L_0}(a) \\ \textit{i.e.} \quad -a^4 V_{L_+}(a) + a^{-4} V_{L_-}(a) &= (a^2 - a^{-2}) V_{L_0}(a) \\ \textit{i.e.} \quad a^4 V_{L_+}(a) - a^{-4} V_{L_-}(a) &= (a^{-2} - a^2) V_{L_0}(a) \end{aligned}$$

We do have the same *skein* relation.

□

Proposition 2.4.39: The Jones polynomial satisfies the following properties:

1. Let L,L' be two links.
   Then $J_{L \sqcup L'}(X) = \frac{1-X}{\sqrt{X}} \cdot J_L(X) \cdot J_{L'}(X)$
2. Let J and K be two knots.
   Then $J_{J\#K}(X) = J_J(X) \cdot J_K(X)$

Proof of proposition 2.4.39: We shall prove these properties later in this paragraph.

Unfortunately, whether it be the Alexander, the Conway or the Jones polynomial, none of them is capable of distinguishing the Conway knot from the Kinoshita-Terasaka knot: they are not complete invariants. There is therefore a need to refine the polynomial invariants still further.

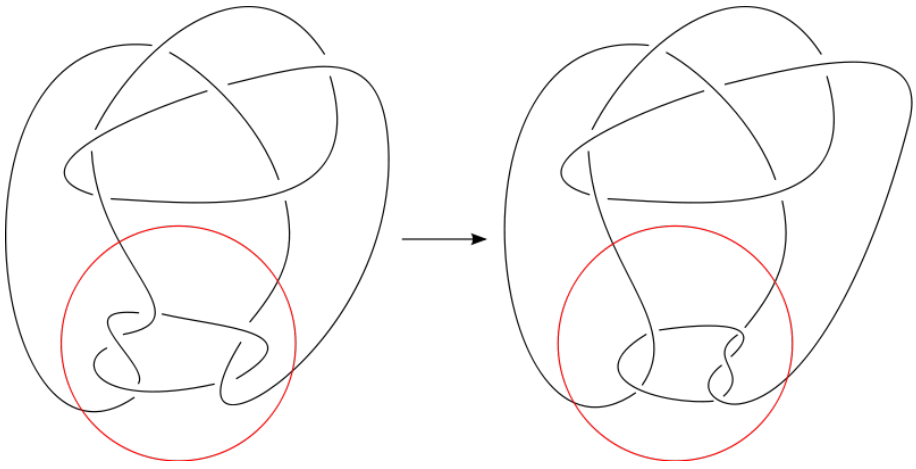


Figure 39: Conway knot (left) and Kinoshita-Terasaka knot (right).

The time has now come to present the HOMFLY-PT polynomial. It was created by the joint effort of 6 researchers, each of whom left their initial in its name: Hoste, Ocneanu, Millett, Freyd, Lickorish, Yetter.

The suffix "PT" comes from the researchers Przytycki and Traczyk, who discovered the polynomial independently of the first 6.

Definition 2.4.40 (HOMFLY-PT polynomial): The two-variable HOMFLY-PT polynomial $P_L(v,z) \in \mathbb{Z}[v^{\pm 1}, z]$ is defined by the following 3 properties:

- $P_{\cdot}$ is a link invariant.
- $P_A(v,z) = 1$
- For every Conway triple (L_+, L_-, L_0), we have:

$$\frac{1}{v} \cdot P_{L_+}(v,z) - v \cdot P_{L_-}(v,z) = z \cdot P_{L_0}(v,z)$$

Proposition 2.4.41 The HOMFLY-PT polynomial satisfies:

i) Let L,L' be two links.
Then $P_{L \sqcup L'}(v,z) = \frac{v^{-1}-v}{z} \cdot P_L(v,z) \cdot P_{L'}(v,z)$

ii) Let J and K be two knots.
Then $P_{J\#K}(v,z) = P_J(v,z) \cdot P_K(v,z)$

iii) For every link L, we have: $\Delta_L(X) = P_L(1, \sqrt{X} - \frac{1}{\sqrt{X}})$

iv) For every link L, we have: $J_L(X) = P_L(X, \sqrt{X} - \frac{1}{\sqrt{X}})$

Proof of proposition 2.4.41: We *accept without proof* the results i) and ii).
Let us prove iii). As before, it suffices to check that these two polynomials have the same image of the unknot (which they do) and that they have the same *skein* relation in order to reach the equality.
Let (L_+, L_-, L_0) be a Conway triple. Set $Y = \sqrt{X} - \frac{1}{\sqrt{X}}$.
Let us rewrite the *skein* relation of the HOMFLY-PT polynomial with $(v,z) = (1,Y)$. It follows that:

$$\frac{1}{1} \cdot P_{L_+}(1,Y) - 1 \cdot P_{L_-}(1,Y) = (\sqrt{X} - \frac{1}{\sqrt{X}}) \cdot P_{L_0}(1,Y)$$
$$\textit{i.e.} \quad P_{L_+}(1,Y) - P_{L_-}(1,Y) = (\sqrt{X} - \frac{1}{\sqrt{X}}) \cdot P_{L_0}(1,Y)$$

Finally, $P_{\cdot}(1,Y)$ does satisfy the same *skein* relation as $\Delta_{\cdot}(X)$. They are therefore equal.
A symmetric argument proves iv).

Thanks to these 2 properties of the HOMFLY-PT polynomial, and to the bridge between this polynomial and Jones's, it is possible to prove the properties of the Jones polynomial stated in proposition 2.4.39.

Let L,L' be two links. Let us show that we do have $J_{L \sqcup L'}(X) = \frac{1-X}{\sqrt{X}} \cdot J_L(X) \cdot J_{L'}(X)$.

Indeed, for $(v,z) = (X, \sqrt{X} - \frac{1}{\sqrt{X}})$, we have:

$$\begin{aligned}
\frac{v^{-1} - v}{z} &= \frac{\frac{1}{X} - X}{\sqrt{X} - \frac{1}{\sqrt{X}}} \\
&= -\frac{(X - \frac{1}{X}) \cdot (\sqrt{X} + \frac{1}{\sqrt{X}})}{X - \frac{1}{X}} \\
&= \frac{1}{\sqrt{X}} \cdot (-X + 1) \\
&= \frac{1 - X}{\sqrt{X}}
\end{aligned}$$

Then finally,

$$P_{L \sqcup L'}(v,z) = \frac{v^{-1} - v}{z} \cdot P_L(v,z) \cdot P_{L'}(v,z)$$

Therefore

$$\begin{aligned}
J_{L \sqcup L'}(X) = P_{L \sqcup L'}(X, \sqrt{X} - \frac{1}{\sqrt{X}}) &= (\frac{1 - X}{\sqrt{X}}) \cdot P_L(X, \sqrt{X} - \frac{1}{\sqrt{X}}) \cdot P_{L'}(X, \sqrt{X} - \frac{1}{\sqrt{X}}) \\
&= (\frac{1 - X}{\sqrt{X}}) \cdot J_L(X) \cdot J_{L'}(X)
\end{aligned}$$

Let J and K be two knots. Let us show that we do have $J_{J\#K}(X) = J_J(X) \cdot J_K(X)$.
Indeed, we have:

$$\begin{aligned}
J_{J\#K}(X) &= P_{J\#K}(X, \sqrt{X} - \frac{1}{\sqrt{X}}) \\
&= P_J(X, \sqrt{X} - \frac{1}{\sqrt{X}}) \cdot P_K(X, \sqrt{X} - \frac{1}{\sqrt{X}}) \\
&= J_J(X) \cdot J_K(X)
\end{aligned}$$

□

Example 2.4.42: The HOMFLY-PT polynomial of the trefoil knot.

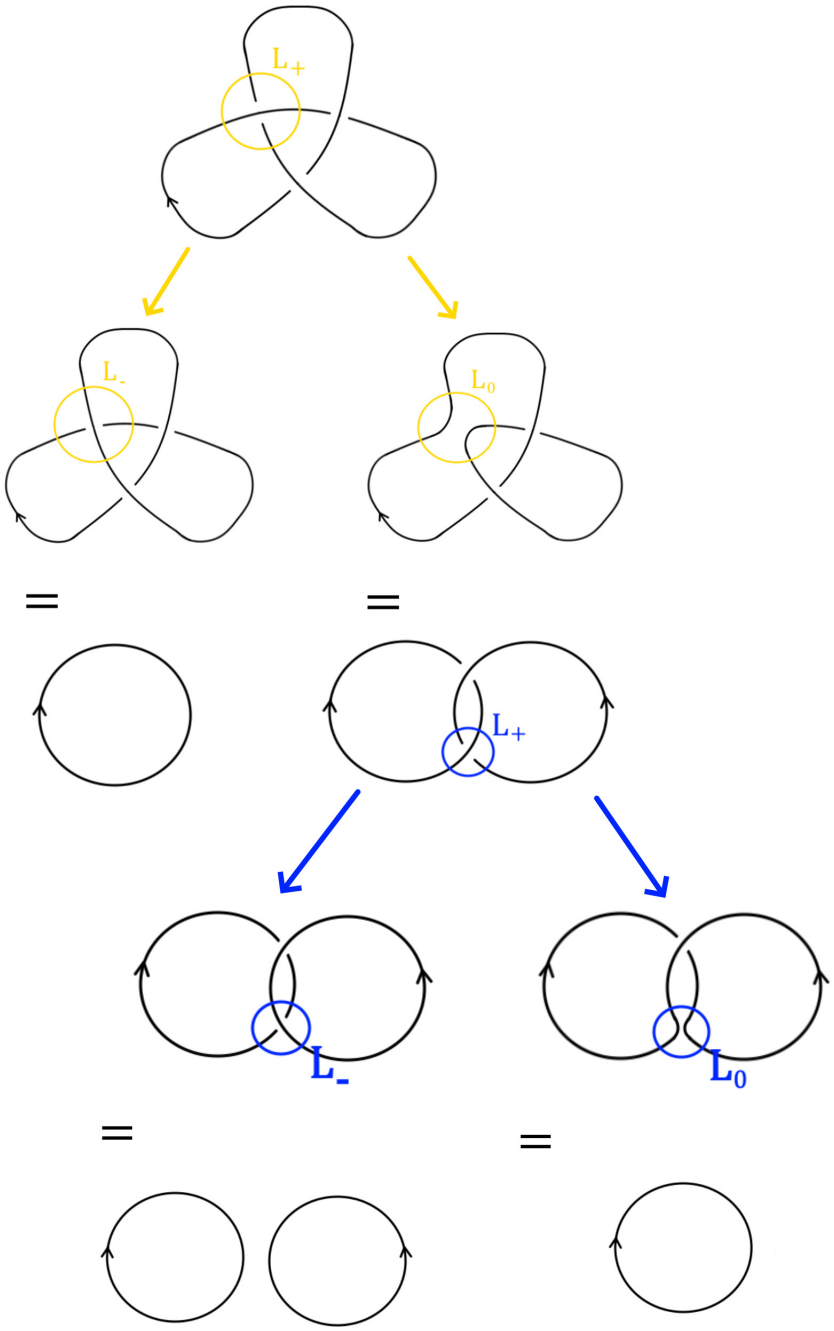


Figure 40: Example of a resolving tree for the trefoil knot T_2.

We shall here use the resolving tree proposed in figure 40 to compute $P_{T_2}(v,z)$. We number the various diagrams obtained from top to bottom and from left to right: from D_1 to D_5. We first consider the yellow triple:

$$\begin{aligned}\frac{1}{v}\cdot P_{D_1}(v,z) - v\cdot P_{D_2}(v,z) = z\cdot P_{D3}(v,z) \Leftrightarrow P_{D_1}(v,z) &= v\cdot(v\cdot P_{D_2}(v,z) + z\cdot P_{D3}(v,z)) \\ &= v^2\cdot P_A(v,z) + vz\cdot P_{D_3}(v,z) \\ &= v^2 + vz\cdot P_{D_3}(v,z)\end{aligned}$$

Then the blue triple:

$$\begin{aligned}\frac{1}{v}\cdot P_{D_3}(v,z) - v\cdot P_{D_4}(v,z) = z\cdot P_{D5}(v,z) \Leftrightarrow P_{D_3}(v,z) &= v\cdot(v\cdot P_{D_4}(v,z) + z\cdot P_{D5}(v,z)) \\ &= v^2\cdot P_{A\sqcup A}(v,z) + vz\cdot P_A(v,z) \\ &= v^2\cdot\frac{v^{-1}-v}{z}\cdot P_A(v,z)\cdot P_A(v,z) + vz \\ &= \frac{v-v^3}{z} + vz\end{aligned}$$

Finally, it follows that:

$$\begin{aligned}P_{T_2}(v,z) = P_{D_1}(v,z) &= v^2 + vz\cdot(\frac{v-v^3}{z} + vz) \\ &= 2v^2 - v^4 + v^2z^2\end{aligned}$$

All in all rather discouragingly, the HOMFLY-PT polynomial is not able to distinguish the Conway knot from the Kinoshita-Terasaka knot either: after 4 different polynomial attempts, none manages to provide a complete invariant of links, or even of knots! Moreover, many mathematicians have seized on the possibilities offered by *skein* relations and have manipulated them in order to obtain ever finer polynomial invariants. But none has succeeded in determining a complete one.
And with good reason: there exists a rather powerful theorem proved by Taizo Kanenobu that exposes the limits of the polynomial approach.

Theorem 2.4.43 (Incompleteness of knot polynomials) (stated without proof): There exist infinitely many distinct knots that have the same polynomial invariant, provided the latter can be defined by a *skein* relation.

Faced with this relative failure of the polynomial invariants, we must change perspective: polynomials have this surprising and uncomfortable feature that they move away very quickly from the *topological* nature of the problem, in order to reduce it immediately to a purely algebraic problem. While it is certain that algebraic tools are very effective, they must nonetheless be tied back to a topological foundation.

2.5 FROM THE FIRST HOMOTOPY GROUP TO THE KNOT GROUP

In line with what we mentioned above, in this section we propose to change approach by introducing a knot invariant coming from algebraic topology.

2.5.1 NOTIONS OF ALGEBRAIC TOPOLOGY

Let Ω be a topological space (whose construction we admit).

Definition 2.5.1 (Path): A path is a continuous map $\Phi : [0,1] \to \Omega$. We call $\Phi(0)$ the origin (resp. $\Phi(1)$ the endpoint).

Definition 2.5.2 (Path-connectedness): We say that Ω is path-connected if:

$$\forall (x,y) \in \Omega^2, \exists \Phi \in \mathcal{C}^0([0,1],\Omega), [\Phi(0) = x \wedge \Phi(1) = y]$$

Definition 2.5.3 (Loop): A loop based at x is a path that satisfies $\Phi(0) = \Phi(1) = x$

Definition 2.5.4 (Concatenation): We say that two paths Φ_1 and Φ_2 of Ω are composable if $\Phi_1(1) = \Phi_2(0)$ (or the converse). We can then compose them according to the following law:

$$\Phi = \Phi_1 * \Phi_2 \text{ with } \Phi_{|[0,1/2]} : t \longmapsto \Phi_1(2t) \text{ and } \Phi_{|[1/2,1]} : t \longmapsto \Phi_2(2t-1)$$

Definition 2.5.5 (Inverse): Let Φ be a path of Ω. We call the "inverse path" (and one checks that it is indeed a path) the map $\overline{\Phi} : [0,1] \to \Omega, t \mapsto \Phi(1-t)$.

Notation: We shall write $\mathcal{C}(\Omega, x_0, x_1)$ for the set of paths of Ω with origin x_0 and endpoint x_1. We shall write $\mathcal{L}(\Omega, x)$ for the set of loops of Ω based at x.

Definition - property 2.5.6 (Homotopy): We say that two paths Φ_0 and Φ_1 of Ω with the same endpoints x_0, x_1 are homotopic if $\exists (\gamma_t)_{0 \leq t \leq 1} \in \mathcal{C}(\Omega, x_0, x_1)^{[0,1]}$ such that:

i) $t \mapsto \gamma_t$ is continuous

ii) $\gamma_0 = \Phi_0, \gamma_1 = \Phi_1$

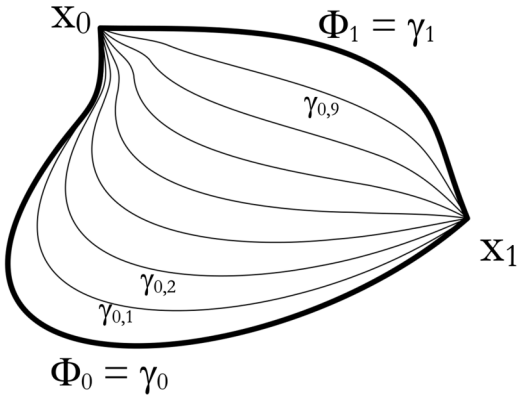


Figure 41: Homotopy between two paths

The binary relation of homotopy defines an equivalence relation on the set of paths of Ω with fixed endpoints, written $\sim$.

Justification of property 2.5.6: Let Φ_0, Φ_1, Φ_2 be three paths with the same endpoints x_0 and x_1.

i) Reflexivity: let us show that $\Phi_0 \sim \Phi_0$.
Candidate: (γ_t) such that $\forall t \in [0,1], \gamma_t := \Phi_0$. As $t \to \gamma_t$ is constant, it is continuous

ii) Symmetry: suppose that $\Phi_0 \sim \Phi_1$ by $(\gamma_t)_{0\leq t\leq 1}$. Let us show that $\Phi_1 \sim \Phi_0$.
Candidate: we consider the family defined by $\forall t \in [0,1], \theta_t := \gamma_{1-t}$.
$t \longmapsto \theta_t$ can be rewritten $t \longmapsto \gamma_t \circ (1-t)$, which is continuous by composition. We moreover obviously have $\gamma_0 = \Phi_1, \gamma_1 = \Phi_0$, whence finally $\Phi_1 \sim \Phi_0$

iii) Transitivity: suppose that $\Phi_0 \sim \Phi_1$ and $\Phi_1 \sim \Phi_2$ by $(\gamma_t)_{0\leq t\leq 1}$ and $(\theta_t)_{0\leq t\leq 1}$ respectively. Let us show that $\Phi_0 \sim \Phi_2$. Candidate: we build the family $(\delta_t)_{0\leq t\leq 1}$ as follows: $\forall t \in [0,1/2], \delta_t := \gamma_{2t}$ and $\forall t \in [1/2,1], \delta_t := \theta_{2t-1}$ This family does satisfy $\delta_0 = \Phi_0, \delta_1 = \Phi_2$, and it remains to show its continuity. It suffices to show that it is continuous at $t = 1/2$, continuity on $[0,1/2[$ and on $]1/2,1]$ being guaranteed by locality. Now, on $[0,1/2]$, δ coincides with γ, which is continuous, so in particular δ is left-continuous at $1/2$, and $\delta_{\frac{1}{2}^-} = \gamma_{\frac{1}{2}^-} = \gamma_{\frac{1}{2}} = \Phi_1$ Similarly, $\delta_{\frac{1}{2}^+} = \theta_{\frac{1}{2}^+} = \theta_{\frac{1}{2}} = \Phi_1$ We deduce that the family δ is continuous, which completes the proof.

□

Property 2.5.7 (Concatenation and homotopy): Homotopy and concatenation are compatible. Let $\gamma_0, \gamma_1 \in \mathcal{C}(\Omega, x_0, x_1)$ and $\eta_0, \eta_1 \in \mathcal{C}(\Omega, x_1, x_2)$ be such that $\gamma_0 \sim \gamma_1$ and $\eta_0 \sim \eta_1$. Set $\varpi_0 := \gamma_0 * \eta_0$ and $\varpi_1 := \gamma_1 * \eta_1$. We then have $\varpi_0 \sim \varpi_1$.

Justification of property 2.5.7: (first of all, the paths are indeed composable).
Let us fix suitable homotopies $(\gamma_t)_{0\leq t\leq 1} \in \mathcal{C}(\Omega, x_0, x_1)^{[0,1]}$ and $(\eta_t)_{0\leq t\leq 1} \in \mathcal{C}(\Omega, x_1, x_2)^{[0,1]}$. Let us show that $\varpi_0, \varpi_1 \in \mathcal{C}(\Omega, x_0, x_2)$ are homotopic.
We set $\forall t \in [0,1], \varpi_t := \gamma_t * \eta_t$. First, by concatenation $(\varpi_t)_{0\leq t\leq 1} \in \mathcal{C}(\Omega, x_1, x_2)^{[0,1]}$. On the other hand, $t \longmapsto \varpi_t$ is continuous. Indeed, let $t_0 \in]0,1]$: $\varpi_{t_0} = \gamma_{t_0} * \eta_{t_0} = \gamma_{t_0^-} * \eta_{t_0^-} = \varpi_{t_0^-}$, by continuity of (γ) and (η), which proves the left-continuity of (ϖ). The same holds for right-continuity.

□

Definition 2.5.8 (Fundamental group): Let $x \in \Omega$; we set $\pi_1(\Omega, x) := \mathcal{L}(\Omega, x)/\sim$. $(\pi_1(\Omega, x), *)$ is a group, called the first homotopy group (or fundamental group) of Ω at x.

Justification of definition 2.5.8: Let us briefly show that $\pi_1(\Omega, x)$ is indeed a group:

i) By property 2.5.7, $*$ passes to the quotient on $\pi_1(\Omega, x)$, and does define a binary operation, by compatibility between concatenation and homotopy. It is immediate that $*$ is associative.

ii) We may consider the identity element defined as the class of the map Φ_0 that satisfies: $\forall t \in [0,1], \Phi_0(t) = x$. Let $\Phi \in \mathcal{L}(\Omega, x)$, and let us show that $[\Phi_0 * \Phi] = [\Phi]$, i.e. let us show that $\Phi_0 * \Phi \sim \Phi$. Consider the family built as follows:
$\forall t \in [0, \frac{1-t}{2}], \gamma_t := \Phi_0$ and $\forall t \in [\frac{1-t}{2}, 1], \gamma_t := \Phi$.
We do have $\gamma_0 = \Phi_0 * \Phi$ by definition 2.5.4, and $\gamma_1 = \Phi$. Finally $t \longmapsto \gamma_t$ is indeed continuous. One shows in the same way that $\forall \Phi \in \mathcal{L}(\Omega, x), [\Phi * \Phi_0] = [\Phi]$.

iii) Let $\Phi \in \mathcal{L}(\Omega, x)$ and let us show that $\overline{\Phi} * \Phi \sim \Phi_0$ (symmetrically one will show that $\Phi * \overline{\Phi} \sim \Phi_0$), which will show that $[\Phi] * [\overline{\Phi}] = [\overline{\Phi}] * [\Phi] = [\Phi_0]$.
Consider the family built as follows:
$\forall t \in [0, \frac{1-t}{2}], \gamma_t := \overline{\Phi}$, $\forall t \in [\frac{1-t}{2}, 1-t], \gamma_t := \Phi$, and $\forall t \in [1-t, 1], \gamma_t := \Phi_0$.
We do have $\gamma_0 = \overline{\Phi} * \Phi$ by definition 2.5.4, and $\gamma_1 = \Phi_0$. Moreover $t \longmapsto \gamma_t$ is indeed continuous.

□

Theorem 2.5.9: Suppose that Ω is path-connected. We may then write, with a slight abuse of notation, $\pi_1(\Omega)$ because:

$$\forall (p,q) \in \Omega^2, \pi_1(\Omega,p) \cong \pi_1(\Omega,q).$$

Proof of theorem 2.5.9: Let $(p,q) \in \Omega^2$. Let us fix γ a path from p to q (legitimate by path-connectedness). Let us show that the following map is an isomorphism:

$$\begin{array}{ccccc} \aleph_\gamma & : & \pi_1(\Omega,q) & \longrightarrow & \pi_1(\Omega,p) \\ & & [\eta] & \longmapsto & [\gamma * \eta * \overline{\gamma}] \end{array}$$

i) $\aleph_\gamma$ is well defined: let $[\eta] \in \pi_1(\Omega,q)$. $\gamma * \eta * \overline{\gamma} \in \mathcal{L}(\Omega,p)$ so $\aleph_\gamma([\eta]) \in \pi_1(\Omega,p)$. Moreover, by compatibility between homotopy and concatenation, $\aleph_\gamma$ does not depend on the choice of the representative.

ii) Let us check that it is a group morphism. Let $[\eta],[\epsilon] \in \pi_1(\Omega,q)$.
$\aleph_\gamma([\eta]*[\epsilon]) = \aleph_\gamma([\eta*\epsilon]) = [\gamma]*[\eta*\epsilon]*[\overline{\gamma}] = [\gamma]*[\eta]*[\overline{\gamma}]*[\gamma]*[\epsilon]*[\overline{\gamma}] = \aleph_\gamma([\eta]) * \aleph_\gamma([\epsilon])$.

iii) Consider $\begin{array}{ccccc} \aleph_{\overline{\gamma}} & : & \pi_1(\Omega,q) & \longrightarrow & \pi_1(\Omega,p) \\ & & [\eta] & \longmapsto & [\overline{\gamma}*\eta*\gamma] \end{array}$. $\forall [\eta] \in \pi_1(\Omega,q), \aleph_{\overline{\gamma}} \circ \aleph_\gamma([\eta]) = [\overline{\gamma}*\gamma*\eta*\overline{\gamma}*\gamma] = [\eta]$.
Therefore $\aleph_{\overline{\gamma}} \circ \aleph_\gamma = Id_{\pi_1(\Omega,q)}$. Similarly $\aleph_\gamma \circ \aleph_{\overline{\gamma}} = Id_{\pi_1(\Omega,p)}$. Therefore $\aleph_\gamma$ is indeed an isomorphism.

□

Property 2.5.10: $\pi_1(\mathcal{S}^1) = \mathbb{Z}$.

Proof of property 2.5.10: As $\mathcal{S}^1$ is path-connected, we reduce to computing $\pi_1(\mathcal{S}^1,1)$.

Preliminary lifting lemma: Let $\gamma \in \mathcal{L}(\mathcal{S}^1)$. Then $\exists ! \tilde{\gamma} \in \mathcal{C}^0([0,1],\mathbb{R}), e^{2i\pi\tilde{\gamma}} = \gamma$ and $\tilde{\gamma}(0) = 0$.

We may therefore define a map $\begin{array}{ccccc} \partial & : & \mathcal{L}(\mathcal{S}^1) & \longrightarrow & \mathbb{R} \\ & & \gamma & \longmapsto & \tilde{\gamma}(1) \end{array}$ where $\tilde{\gamma}$ is the unique lift of γ, guaranteed by the preliminary lemma.
Note that, as γ is a loop, $e^{2i\pi\tilde{\gamma}(1)} = \gamma(1) = 1 = \gamma(0) = e^{2i\pi\tilde{\gamma}(0)} = 1$, so $\tilde{\gamma}(1) \in \mathbb{Z}$.
Let us show that we can deduce from this the existence of an isomorphism $\begin{array}{ccccc} \partial & : & \pi_1(\mathcal{S}^1) & \longrightarrow & \mathbb{Z} \\ & & [\gamma] & \longmapsto & \partial(\gamma) \end{array}$.

i) Let us show that ∂ is well defined.

Let γ_0 and γ_1 be two homotopic loops, by the homotopy (γ_t). The map $(s,t) \longmapsto \gamma_t(s)$ is continuous by definition 2.5.6. For every $t \in [0,1]$, we have the associated lift $\tilde{\gamma}_t$.
We take for granted that the map $(s,t) \longmapsto \tilde{\gamma}_t(s)$ is also continuous (homotopy lifting theorem).
So in particular $t \longmapsto \tilde{\gamma}(1) = \partial(\gamma_t)$ is continuous. Now this map takes values in $\mathbb{Z}$: it is therefore constant. Therefore $\partial(\gamma_0) = \partial(\gamma_1)$.

Summary: $\forall \gamma_0, \gamma_1 \in \mathcal{L}(\mathcal{S}^1,1), [\gamma_0 \sim \gamma_1 \implies \partial(\gamma_0) = \partial(\gamma_1)]$. Therefore ∂ is well defined on $\pi_1(\mathcal{S}^1)$.

ii) Let us show that ∂ is a group morphism.

Let $[\gamma_1],[\gamma_2] \in \pi_1(\mathcal{S}^1)$.
Let us show that $\partial([\gamma_1]*[\gamma_2]) = \partial([\gamma_1]) + \partial([\gamma_2])$. Let us show that $\partial(\gamma_1*\gamma_2) = \partial(\gamma_1) + \partial(\gamma_2)$.
Let us fix $\tilde{\gamma}_1, \tilde{\gamma}_2$ the respective unique lifts of γ_1, γ_2 satisfying $\tilde{\gamma}_1(0) = \tilde{\gamma}_1(0) = 0$.
Set $\begin{array}{ccccc} \tilde{\gamma} & : & [0,1] & \longrightarrow & \mathbb{R} \\ & & t & \longmapsto & \begin{cases} \tilde{\gamma}_1(2t) & \text{if } t \leq 1/2 \\ \tilde{\gamma}_2(2t-1) + \tilde{\gamma}_1(1) & \text{otherwise} \end{cases} \end{array}$
By construction, $\tilde{\gamma}$ is indeed continuous, $e^{2i\pi\tilde{\gamma}} = \gamma_1 * \gamma_2$, and $\tilde{\gamma}(0) = \tilde{\gamma}_1(0) = 0$. It is therefore the unique lift of $\gamma_1 * \gamma_2$ satisfying the properties of the preliminary lifting lemma.
So by definition of ∂, we have $\partial(\gamma_1*\gamma_2) = \tilde{\gamma}(1) = \tilde{\gamma}_2(1) + \tilde{\gamma}_1(1) = \partial(\gamma_1) + \partial(\gamma_2)$.

Summary: $\partial : \pi_1(\mathcal{S}^1) \longrightarrow \mathbb{Z}$ is a group morphism.

iii) Let us show that ∂ is injective.

Let $[\gamma] \in \mathrm{Ker}(\partial)$. Let us show that $\gamma \sim l_1$, the constant loop based at 1.
Consider the lift $\tilde{\gamma}$ of γ. We have $\tilde{\gamma_1}(1) = \partial(\gamma) = 0 = \tilde{\gamma}(0)$.
Set $\forall t \in [0,1], \gamma_t := e^{2i\pi t\tilde{\gamma}}$. The map $(t,s) \longmapsto \gamma_t(s)$ is continuous. Moreover: $\gamma_1 = e^{2i\pi\tilde{\gamma}} = \gamma$ and $\gamma_0 = l_1$. Therefore $\gamma \sim l_1$.

Summary: $\mathrm{Ker}(\partial) = \{l_1\}$.

iv) Let us show that ∂ is surjective.

Let $n \in \mathbb{Z}$. We set $\forall t \in [0,1], \gamma t := e^{2i\pi nt}$ which is indeed a loop.
Note that $t \longmapsto \tilde{\gamma}(t) = nt$ is a lift of γ satisfying $\tilde{\gamma}(0) = 0$. Therefore $\partial(\gamma) = \tilde{\gamma}(1) = n$.

Summary: $\mathrm{Im}(\partial) = \mathbb{Z}$.

Conclusion: $\pi_1(\mathcal{S}^1) \cong \mathbb{Z}$.

□

Definition 2.5.11 (Simple connectedness): A path-connected space Ω is said to be simply connected (or 1-connected) when every loop is homotopic to the trivial loop. In this case, $\pi_1(\Omega) = \{1\}$.

A cultural remark from category theory: a covariant functor from the category $\mathcal{C}$ to the category $\mathcal{C}'$ is a map $F : \mathcal{C} \to \mathcal{C}'$, which to every object $X \in \mathcal{C}$ associates an object $F(X) \in \mathcal{C}'$, and to every morphism $\phi : \mathcal{C} \to \mathcal{C}$ a morphism $\phi_* : \mathcal{C}' \to \mathcal{C}'$. As the following definition illustrates, Π_1 is a covariant functor from Top_* (the category of pointed topological spaces) to Grp (the category of groups).

Definition 2.5.12 (Induced morphism): Let Ω and Λ be two topological spaces and $\phi : \Omega \to \Lambda$ a continuous map. It induces a group morphism (through the application of the functor Π_1):

$$\begin{array}{ccccc} \phi_* & : & \pi_1(\Omega, x_0) & \longrightarrow & \pi_1(\Lambda, \phi(x_0)) \\ & & [f] & \longmapsto & [\phi f] \end{array}$$

so that the following diagram commutes:

$$\begin{array}{ccc} (\Omega, x_0) & \xrightarrow{\Pi_1} & \pi_1(\Omega, x_0) \\ \phi \downarrow & & \downarrow \phi_* \\ (\Lambda, \phi(x_0)) & \xrightarrow{\Pi_1} & \pi_1(\Lambda, \phi(x_0)) \end{array}$$

If moreover we have a continuous map $\psi : \Lambda \to \Omega$, then $(\phi \circ \psi)_* = \phi_* \circ \psi_*$ and $(Id_\Omega)_* = Id_{\pi_1(\Omega, x_0)}$

Theorem 2.5.13 (Homeomorphism theorem): Let Ω and Λ be two homeomorphic topological spaces (which we shall take here to be path-connected). Then $\pi_1(\Omega) \cong \pi_1(\Lambda)$.

Proof of theorem 2.5.13: Let us fix $f : \Omega \to \Lambda$ and $g : \Lambda \to \Omega$ two mutually inverse homeomorphisms. As f and g are inverse to each other, $f \circ g = Id_\Lambda$ and $g \circ f = Id_\Omega$. Consider $f_* : \pi_1(\Omega) \to \pi_1(\Lambda)$ and $g_* : \pi_1(\Lambda) \to \pi_1(\Omega)$. It therefore follows that

$$\begin{cases} f_* \circ g_* = (f \circ g)_* = (Id_\Lambda)_* = Id_{\pi_1(\Lambda)} \\ g_* \circ f_* = (g \circ f)_* = (Id_\Omega)_* = Id_{\pi_1(\Omega)} \end{cases}$$

Therefore f_* and g_* are inverse to each other, hence are group isomorphisms, whence $\pi_1(\Omega) \cong \pi_1(\Lambda)$.

□

2.5.2 TOOLS FOR COMPUTING FUNDAMENTAL GROUPS

With a view to constructing a knot invariant, we must equip ourselves with practical tools for computing the fundamental group of a space.

Definition 2.5.14 (Free product of groups): Let G_1 and G_2 be two groups. We define their free product, written $G_1 \star G_2$, as the set of words over G_1, G_2, that is, the elements of the form $a_1 a_2 ... a_n$, where $\forall k \in \{1, ..., n-1\}$, $[a_k \in G_1(\text{resp. } G_2) \Rightarrow a_{k+1} \in G_2(\text{resp. } G_1)]$. Equipped with a natural concatenation law of composition, $G_1 * G_2$ is a group.

Remark 2.5.15: If respective presentations of G_1 and G_2 are $G_1 = \langle R_{G_1} \mid S_{G_1} \rangle$ and $G_2 = \langle R_{G_2} \mid S_{G_2} \rangle$, then a presentation of their free product is: $G_1 * G_2 = \langle R_{G_1} \cup R_{G_2} \mid S_{G_1} \cup S_{G_2} \rangle$.

Example 2.5.16: $\mathbb{Z}/2\mathbb{Z} = \langle a \mid a^2 = 1 \rangle$, $\mathbb{Z}/3\mathbb{Z} = \langle b \mid b^3 = 1 \rangle$, so $\mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/3\mathbb{Z} = \langle a, b \mid a^2 = 1 = b^3 \rangle$.

Definition 2.5.17 (Amalgamated product): Let G, G_1, G_2 be groups, and $\phi_1 : G \to G_1$ and $\phi_2 : G \to G_2$ group morphisms. Consider the morphisms $\theta_1 : G_1 \to G_1 \star G_2$ and $\theta_2 : G_2 \to G_1 \star G_2$. Set $N := min\{M \trianglelefteq G_1 \star G_2 \mid \forall g \in G, (\theta_1(\phi_1(g)))(\theta_1(\phi_1(g)))^{-1} \in M\}$. Then $G_1 \star_G G_2 := (G_1 \star G_2)/N$ is called the amalgamated product of $G1$ and $G2$ over G.

Remark 2.5.18: Let us illustrate this definition. Let $G_1 = \langle R_{G_1} \mid S_{G_1} \rangle$ and $G_2 = \langle R_{G_2} \mid S_{G_2} \rangle$ be respective presentations of G_1 and G_2. Let $\Phi : G \to G_1$ and $\Psi : G \to G_2$ be injective morphisms.
Then a presentation of the amalgamated product over G is: $\langle R_{G_1} \cup R_{G_2} \mid S_{G_1} \cup S_{G_2} \cup [\forall x \in G, \Phi(x) = \Psi(x)] \rangle$.
Intuitively, we form the free product $G_1 * G_2$ and then quotient by the relation $[\forall x \in G, \Phi(x) = \Psi(x)]$.

Example 2.5.19: With this remark, it follows that the free product is in fact the amalgamated product over the trivial group. Indeed, $\forall x \in \{1\}, \Phi(x) = \Psi(x) = 1_{G_1 * G_2}$, so the presentation of the amalgamated product simply gives $\langle R_{G_1} \cup R_{G_2} \mid S_{G_1} \cup S_{G_2} \rangle$, whence $G_1 * G_2 = G_1 *_{\{1\}} G_2$.

Theorem 2.5.20 (Seifert - Van Kampen) (stated without proof): Suppose that $\Omega = U_1 \cup U_2$ with U_1, U_2 two path-connected open sets and $U_0 := U_1 \cap U_2$ path-connected. Let $x \in U_0$. Then:

$$\pi_1(\Omega, x) = \pi_1(U_1, x) \star_{\pi_1(U_0, x)} \pi_1(U_2, x)$$

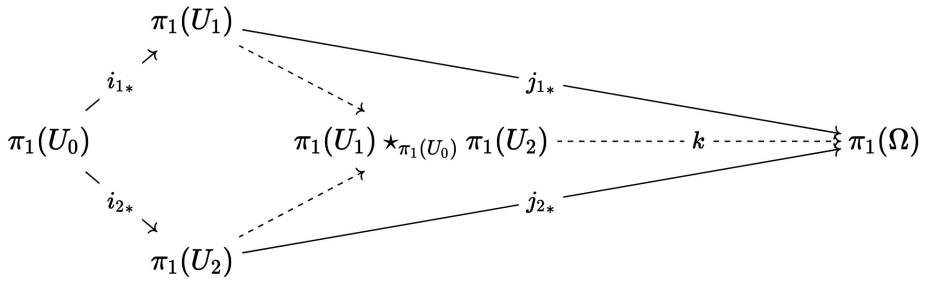
Informal explanation of theorem 2.5.20: We do not specify the morphisms with which we amalgamate, because we have the following canonical injections:

$$\begin{array}{ccc} U_0 & \xrightarrow{i_2} & U_2 \\ \downarrow{\scriptstyle i_1} & \searrow & \downarrow{\scriptstyle j_2} \\ U_1 & \xrightarrow{j_1} & \Omega \end{array}$$

By applying the functor Π_1 (in the sense of definition 2.5.12), the following diagram is always commutative

$$\begin{array}{ccc} \pi_1(U_0) & \xrightarrow{i_{2*}} & \pi_1(U_2) \\ \downarrow{\scriptstyle i_{1*}} & \searrow & \downarrow{\scriptstyle j_{2*}} \\ \pi_1(U_1) & \xrightarrow{j_{1*}} & \pi_1(\Omega) \end{array}$$

Now, every point $x \in U_0$ is identified in U_2 and in U_1 by the maps i_1, i_2: we take for granted that this property is inherited through the application of Π_1, which corresponds to the following amalgamated product:



This commutative diagram brings out the forced identification (amalgamated product) of the elements of $\pi_1(U_0)$ in $\pi_1(U_1)$ and $\pi_1(U_2)$, corroborating remark 2.5.18, and asserts that k is an isomorphism.

Definition - property 2.5.21 (Deformation retract): Let $X \subset \Omega$, which we assume moreover to be path-connected. A deformation retraction of Ω onto X is a family of continuous maps $\xi_t : \Omega \to \Omega$ for $(t \in [0,1])$ such that:

i) $\xi_0 = Id_\Omega$

ii) $\xi_1(\Omega) = X$

iii) $\forall t \in [0,1], \xi_{|X} = Id_X$

iv) $(t,x) \longmapsto \xi_t(x)$ is continuous

In this case, we have $\pi_1(\Omega) = \pi_1(X)$.

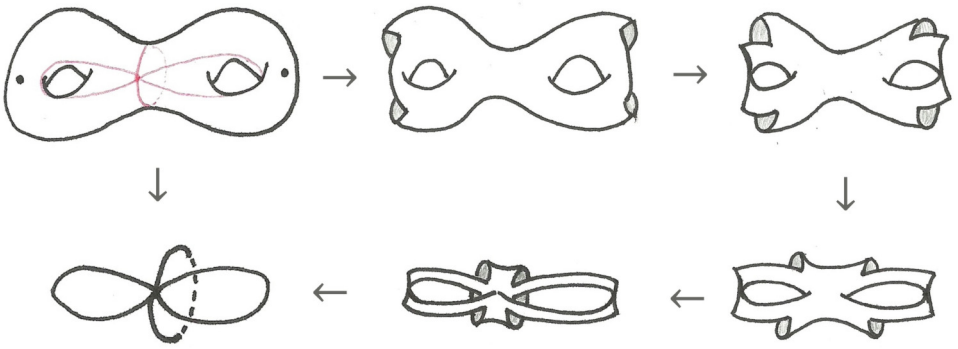


Figure 42: Deformation retraction of the twice-punctured $\mathbb{T}^2 \# \mathbb{T}^2$ onto its core

Proof of property 2.5.21:

Let us show that the canonical injection $\iota : \pi_1(X) \to \pi_1(\Omega)$ is an isomorphism.

In what follows, for γ a loop of X, we shall denote by $[\gamma]_X$ its class in $\pi_1(X)$, and by $[\gamma]_\Omega = \iota([\gamma]_X)$ its class in $\pi_1(\Omega)$. We shall also write $[\gamma]_\Omega$ for a plain loop of Ω, when no confusion is possible.

i) Let us show that ι is a morphism.

It is obvious that every loop of X can be seen as a loop of Ω, which gives ι its canonical character. It is obvious that this injection is compatible with homotopy and concatenation.

ii) Let us show that ι is surjective.

Let $[\gamma]_\Omega \in \pi_1(\Omega)$; let us show that $\exists \mu \in \mathcal{L}(X), [\gamma]_\Omega = [\mu]_\Omega$. We set $\forall t \in [0,1], G_t := \xi_t \circ \gamma$.

We have $G_0 = \xi_0 \circ \gamma = \gamma$ because $\xi_0 = Id_\Omega$. And G_1 takes values in X because ξ_1 does as well: G_1 is therefore a loop of X, which we write μ. The family (G_t) thus built constitutes a homotopy between γ and μ in Ω. Therefore $[\gamma]_\Omega = [\mu]_\Omega = \iota([\mu]_X)$, whence the surjectivity of ι.

iii) Let us show that ι is injective.

Let $[\gamma]_X$ be such that $\iota([\gamma]_X) = [\gamma]_\Omega = 1_{\pi_1(\Omega)}$ (the class of the trivial loop in Ω). We therefore have a homotopy between γ and 1_Ω (the trivial loop in Ω) in Ω, which we write (G_t). We write $r := \xi_1$, which constitutes the retraction (projection of Ω into X). Set $\forall t \in [0,1], H_t := r \circ G_t$. One checks that (H_t) defines a homotopy between γ and 1_X in X. Therefore $[\gamma]_X = 1_{\pi_1(X)}$, whence the injectivity of ι.

In conclusion, $\pi_1(\Omega) \cong \pi_1(X)$

□

Example 2.5.22 (1-connectedness of the 2-sphere): Without loss of generality, we take the unit 2-sphere.

$\mathcal{S}^2 = \mathbb{D}^- \cup \mathbb{D}^+$ with $\mathbb{D}^+ := \{(x,y,z) \in \mathcal{S}^2 \mid z > -\frac{1}{4}\}$ and $\mathbb{D}^- := \{(x,y,z) \in \mathcal{S}^2 \mid z < \frac{1}{4}\}$ which are open sets. $\mathbb{D}^- \cap \mathbb{D}^+$ is a path-connected annulus that deformation retracts onto the equator $\mathcal{S}^1$. $\mathbb{D}^-$ and $\mathbb{D}^+$ are also path-connected, so by the Seifert-Van Kampen theorem: $\pi_1(\mathcal{S}^2) = \pi_1(\mathbb{D}^-) \star_{\pi_1(\mathbb{D}^- \cap \mathbb{D}^+)} \pi_1(\mathbb{D}^+)$. Now, $\mathbb{D}^\pm$ is homeomorphic to an open disc that deformation retracts onto its centre, hence has a trivial fundamental group, so: $\pi_1(\mathcal{S}^2) = \{1\} \star_{\pi_1(\mathcal{S}^1)} \{1\} = \{1\}$.

□

Definition 2.5.23 (Wedge sum): Let Ω and Λ be two spaces, and $x_0 \in \Omega,\ y_0 \in \Lambda$. The wedge sum $\Omega \vee \Lambda$ is obtained as the disjoint union $\Omega \sqcup \Lambda$ quotiented by identifying $x_0 = y_0$. We can extend this definition to a family $(\Omega_i)_{i \in I}$ by setting $\bigvee_{i \in I} \Omega_i = \bigsqcup_{i \in I} \Omega_i / \sim$ with $\sim$ the identification of the points $(x_i)_{i \in I}$.

Example 2.5.24: The following figure is obtained as the wedge sum $\mathcal{S}^1 \vee \mathcal{S}^1$:

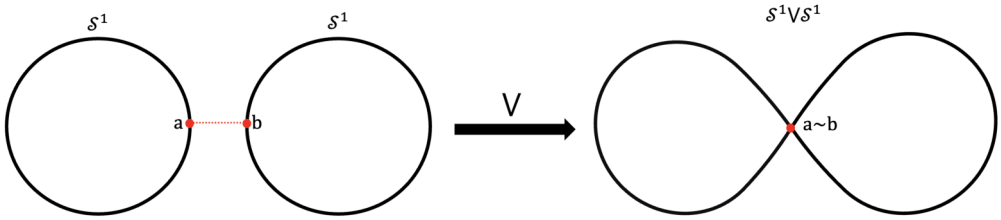


Figure 43: Wedge sum $\mathcal{S}^1 \vee \mathcal{S}^1$ on the lemniscate of Bernoulli

Property 2.5.25 (Group of a wedge sum): Let us take again Ω, Λ disjoint and $x_0 \in \Omega,\ y_0 \in \Lambda$. Suppose that x_0 (resp. y_0) is the deformation retract of an open neighbourhood U_{x_0} (resp. U_{y_0}) in Ω (resp. Λ). Set $A := \Omega \vee U_{y_0}$ and $B := \Lambda \vee U_{x_0}$.

As Ω and Λ are disjoint, A and B are open, and moreover Ω (resp. Λ) is the deformation retract of A (resp. B) by retraction of U_{y_0} onto $\{y_0\} = \{x_0\}$ (resp. U_{x_0} onto $\{y_0\} = \{x_0\}$). And $A \cap B = U_{x_0} \vee U_{y_0}$, which deformation retracts onto the point x_0. Finally, $A \cup B$ deformation retracts onto $\Omega \vee \Lambda$.

Therefore, by deformation retraction, and by the Seifert - Van Kampen theorem: $\pi_1(\Omega \vee \Lambda) = \pi_1(A \cup B) = \pi_1(A) \star_{\pi_1(A \cap B)} \pi_1(B) = \pi_1(\Omega) \star_{\{1\}} \pi_1(\Lambda) = \pi_1(\Omega) \star \pi_1(\Lambda)$, using example 2.5.19 on amalgamation over the trivial group.

□

2.5.3 CONSTRUCTION OF THE KNOT GROUP

Definition 2.5.26 (Knot group): Let K be a knot. We call the "group of K" the fundamental group of its complement: $G(K) := \pi_1(\mathbb{R}^3 - K)$ (the complement is indeed path-connected).

Remark 2.5.27: We may consider that the knot K is a deformation retract of one of its tubular neighbourhoods N, which does not affect its fundamental group by property 2.5.21. In this way we reduce to a slightly more "substantial" topological space. In this case, $G(K) = \pi_1(\mathbb{R}^3 - N)$.

Theorem 2.5.28 (Invariant): We take for granted that the complement is a knot invariant (the result is rather intuitive, since we have considered *ambient* isotopies, which therefore act only locally and leave the complement of the knot globally invariant, up to homeomorphism). We showed earlier that the fundamental group was a topological invariant (theorem 2.5.13), which finally allows us to consider that the knot group is a knot invariant.

In what follows, we propose to determine the knot group of a few knots in order to prove that they are not equivalent in the sense of isotopy. To do so, we shall introduce a generic method for easily obtaining a presentation of the knot group.

Theorem 2.5.29 (Wirtinger presentation): We arbitrarily orient the knot diagram (this does not change the final result), then we distinguish the "elementary components" of the diagram, called arcs in what follows (these are the strokes that are not cut). We then choose a base point x_0 of the complement, and we represent the elementary loops based at x_0 that encircle each component, oriented in such a way that on curving the palm of the right hand around an arc in the direction of the loop, the thumb points in the same direction as the arc.

We must then observe relations on the various loops. Consider a "positive" crossing, for instance between $\alpha_i, \alpha_{i+1}, \alpha_k$. We have the following elementary loops based at x_0: x_i, x_{i+1}, x_k. The following figure illustrates that we have the relation $R_+ : x_k x_i = x_{i+1} x_k$, that is $R_+ : x_{i+1} = x_k x_i {x_k}^{-1}$.

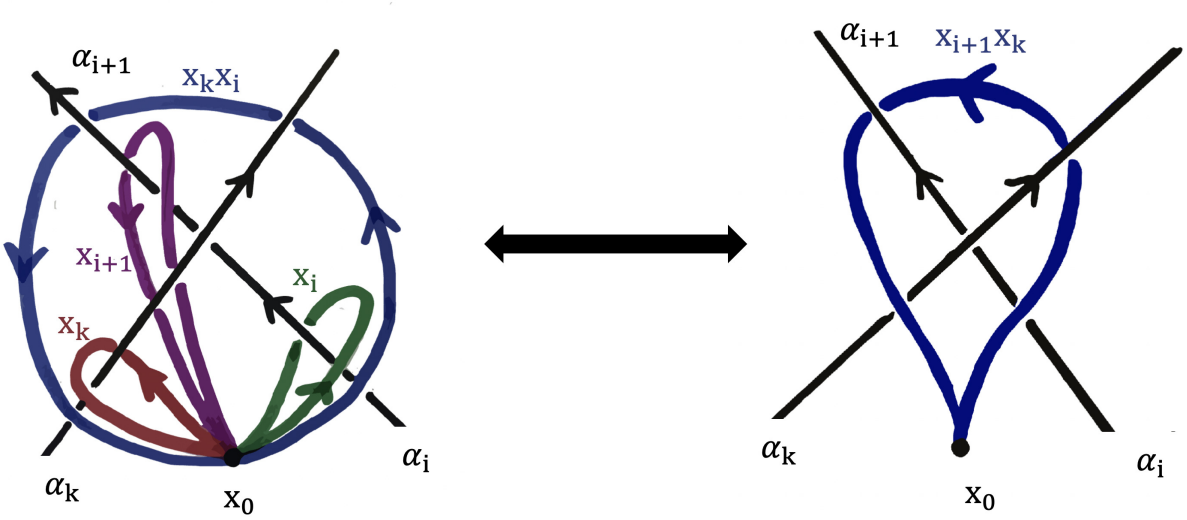


Figure 44: Relations between the loops at a positive crossing

Similarly, at a "negative" crossing, we would have: $R_- : x_{i+1} = {x_k}^{-1} x_i x_k$.

The group of the knot K is therefore generated by the elementary loops based at x_0, which satisfy all the relations at the crossings: $G(K) = \langle x_1, ..., x_n \mid R_-, R_+ \rangle$.

Proof of theorem 2.5.29:

Without loss of generality, and up to isotopy, we may assume that each arc α_i of the knot K lies in the plane $z = 1$, apart from vertical segments at the ends of the arcs that go down to the plane $z = 0$. The final point of the arc α_i can be connected to the initial point of the arc α_{i+1} by a segment β_j in the plane $z = 0$. By virtue of remark 2.5.27, we consider an ϵ-tubular neighbourhood N of the knot K, so that $\mathbb{R}^3 - N$ is a deformation retract of $\mathbb{R}^3 - K$.

This is illustrated in the following figure, where we have represented the tubular neighbourhood of the knot (square here) and distinguished the arcs.

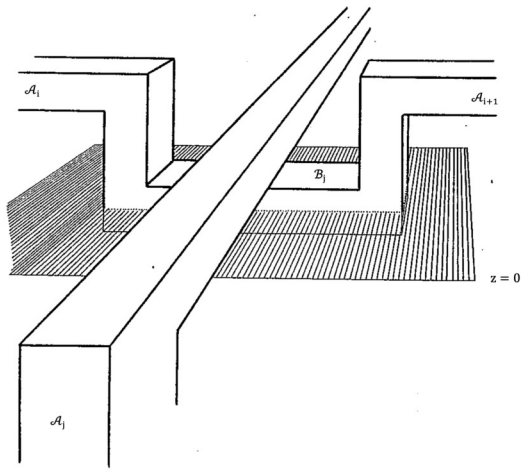


Figure 45: Tubular neighbourhood of the arcs of the knot and the plane $z = 0$

Observe that $\mathbb{R}^3 - K = \mathcal{A} \cup \mathcal{B}$, where we have set $\begin{cases} \mathcal{A} := \{z > 0\} - N \\ \mathcal{B} := \{z < \epsilon/2\} - N \end{cases}$

On the one hand, $\mathcal{A}$ is an open set that retracts onto a wedge of circles $x_1, ..., x_n$, where x_i is a loop passing under the tubular neighbourhood $\mathcal{A}_i$ of the arc α_i, as shown in figure 46. By an immediate induction on property 2.5.24, and using the fact that $\pi_1(\mathcal{S}^1) = \mathbb{Z}$ (Example 2.5.10), we have $\pi_1(\mathcal{A}) = \star_n \mathbb{Z} = \langle x_1, ..., x_n \rangle$.

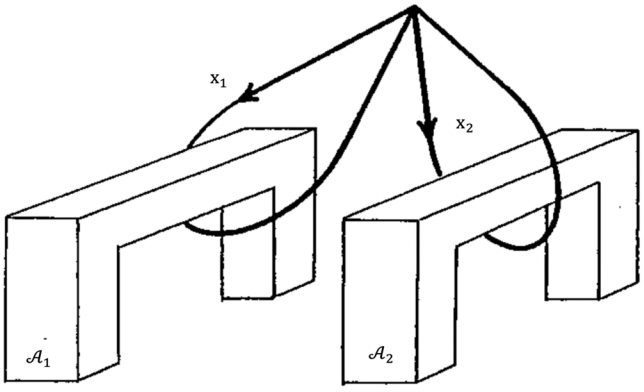


Figure 46: Retraction of $\mathcal{A}$ onto a wedge of circles

On the other hand, $\mathcal{B}$ is an open space with cavities corresponding to the tubular neighbourhoods $\mathcal{B}_j$ of the segments β_j: $\mathcal{B}$ is clearly simply connected, so $\pi_1(\mathcal{B}) = \{1\}$.

Finally $\mathcal{A} \cap \mathcal{B} = \{0 < z < \epsilon/2\} - N$ is a "slab" of thickness $\epsilon/2$ with holes, corresponding to the $\mathcal{B}_j$.

Let g be a generator of $\pi_1(\mathcal{A} \cap \mathcal{B})$. Set $\begin{cases} i : \mathcal{A} \cup \mathcal{B} \hookrightarrow \mathcal{A} \\ j : \mathcal{A} \cup \mathcal{B} \hookrightarrow \mathcal{B} \end{cases}$ the canonical injections, and write i_*, j_* for the associated morphisms in the sense of definition 2.5.12.

One checks that $i_*(g) = x_i x_j^{-1} x_{i+1}^{-1} x_j$ (formally, this is what we did in figure 44) if the crossing is positive, or $i_*(g) = x_i x_j x_{i+1}^{-1} x_j^{-1}$ if it is negative. And as $\pi_1(\mathcal{B}) = \{1\}$, we of course have $j_*(g) = 1$.

Using remark 2.5.18 for the Van Kampen theorem, we obtain the following presentation for the desired amalgamated product:
$\pi_1(\mathcal{A}) \star_{\pi_1(\mathcal{A}\cap\mathcal{B})} \pi_1(\mathcal{B}) = \langle x_1, ..., x_n \mid [\forall g \in \pi_1(\mathcal{A} \cup \mathcal{B}), (x_i x_j^{-1} x_{i+1}^{-1} x_j = 1) \text{ or } (x_i x_j x_{i+1}^{-1} x_j^{-1} = 1)]\rangle$.

This finally translates into $\pi_1(\mathbb{R}^3 - K) = \pi_1(\mathbb{R}^3 - N) = \langle x_1, ..., x_n \mid R_+, R_-\rangle$, where R_+, R_- generically denote all the relations at the crossings, for any triple of concurrent arcs.

NB: To be perfectly rigorous, one would have to prove that the generators g of $\pi_1(\mathcal{A}\cap\mathcal{B})$ describe exactly all the $x_i x_j^{-1} x_{i+1}^{-1} x_j$ or $x_i x_j x_{i+1}^{-1} x_j^{-1}$ under i_*, before asserting that we exhaustively have the relations R_-, R_+, which we take for granted here.

□

Unknot

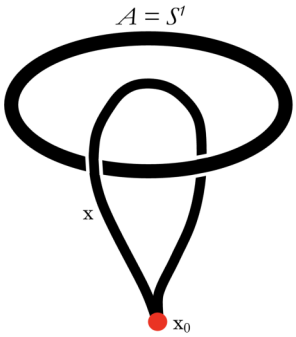


Figure 47: Loops on the complement of the unknot

It is immediate that the Wirtinger presentation of the group of the unknot $G(A)$ is expressed as $G(A) = \langle x \rangle$ with x an arbitrary non-trivial loop based at x_0.
It therefore follows that $G(A) \cong \mathbb{Z}$.

But one can convince oneself of this by going through the Seifert-Van Kampen theorem.

The complement of $A = \mathcal{S}^1$ in $\mathbb{R}^3$ can deformation retract onto a sphere $\mathcal{S}^2$ equipped with one of its diameters d, as in the following figure, which is homeomorphic to the wedge sum $\mathcal{S}^2 \vee \mathcal{S}^1$.

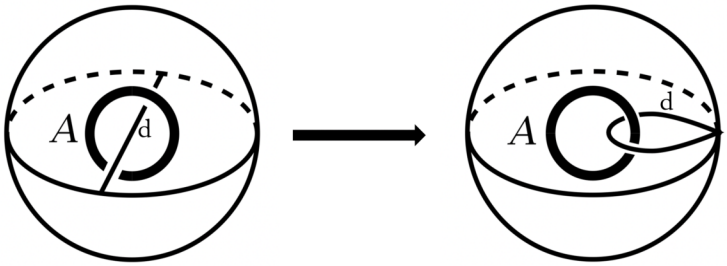


Figure 48: Deformation retraction of the complement of A onto $\mathcal{S}^2 \vee \mathcal{S}^1$

So by properties 2.5.13, 2.5.21, 2.5.25: $\pi_1(\mathbb{R}^3 - \mathcal{S}^1) = \pi_1(\mathcal{S}^2 \vee \mathcal{S}^1) = \pi_1(\mathcal{S}^1) \star \pi_1(\mathcal{S}^2)$.

Now, by example 2.5.22, $\pi_1(\mathcal{S}^2) = \{0\}$. Therefore $\pi_1(\mathbb{R}^3 - \mathcal{S}^1) = \pi_1(\mathcal{S}^1) \star \{0\} = \pi_1(\mathcal{S}^1) = \mathbb{Z}$.

Left trefoil knot

The Wirtinger presentation of the group of the trefoil knot T_1 (figure 2) is expressed as:

$$G(T_1) = \langle x_1, x_2, x_3 \mid x_3 = {x_1}^{-1} x_2 x_1, x_2 = {x_3}^{-1} x_1 x_3, x_1 = {x_2}^{-1} x_3 x_2 \rangle$$

We note that x_1 and x_2 already generate $G(T_1)$, and substituting the first relation into the other two, we have: $\begin{cases} x_2 = ({x_1}^{-1} x_2 x_1)^{-1} x_1 ({x_1}^{-1} x_2 x_1) \\ x_1 = {x_2}^{-1} ({x_1}^{-1} x_2 x_1) x_2 \end{cases}$

That is: $\begin{cases} x_2 x_1 x_2 = x_1 x_2 x_1 \\ x_1 x_2 x_1 = x_2 x_1 x_2 \end{cases}$

It therefore follows that: $G(T_1) = \langle x_1, x_2 \mid x_1 x_2 x_1 = x_2 x_1 x_2 \rangle$.

Set $p = x_1 x_2$ and $q = x_1 x_2 x_1$. We have in particular: $\begin{cases} x_1 = p^{-1} q \\ x_2 = {x_1}^{-1} x_1 x_2 = q^{-1} p^2 \end{cases}$

so p, q generate $G(T_1)$.

The relations are expressed as: $(x_1 x_2 x_1)(x_2 x_1 x_2) = (x_1 x_2)(x_1 x_2)(x_1 x_2)$, that is $q^2 = p^3$.

Thus, $G(T_1) = \langle p, q \mid q^2 = p^3 \rangle$.

Note in particular that $G(T_1)$ and $G(A)$ are not isomorphic, which confirms that the trefoil knot is not trivial.

Figure-eight knot

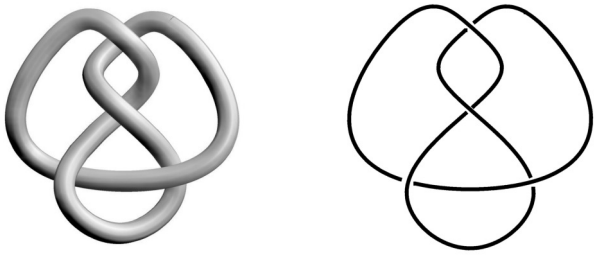


Figure 49: Figure-eight knot and its diagram

The Wirtinger presentation of the group of the figure-eight knot H is:

$$G(H) = \langle a, b, c, d \mid c = a^{-1} b a,\ b = d a d^{-1},\ d = b c b^{-1},\ a = c^{-1} d c \rangle$$

Here too, we note that a and d already generate $G(H)$, and that the last relation follows from the first three, so:

$$G(H) = \langle a, d \mid c = a^{-1}ba,\ b = dad^{-1},\ d = bcb^{-1} \rangle$$

That is:

$$\begin{cases} db = bcb^{-1}b = bc = ba^{-1}ba \\ b = dad^{-1} \end{cases}$$

By substituting the relation $b = dad^{-1}$ into the first one, we obtain: $a^{-1}dad^{-1}a = da^{-1}dad^{-1}$, that is $ada^{-1}da = dad^{-1}ad$.

We finally obtain the following presentation: $G(H) = \langle a, d \mid ada^{-1}da = dad^{-1}ad \rangle$.

Note in particular that $G(H)$ and $G(A)$ are not isomorphic, which confirms that the figure-eight knot is not trivial.

Remark 2.5.30: The trefoil knot and the figure-eight knot are not equivalent.

Justification of remark 2.5.30: Arguing by contradiction, suppose that the knots are isotopic: their groups would have to be isomorphic. Recall that $\mathfrak{S}_3 = \{1, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$

By setting $\Phi(p) = (1\ 2\ 3)$ and $\Phi(q) = (1\ 2)$, we have a surjective morphism $\Phi : G(T_1) \longrightarrow \mathfrak{S}_3$.

Indeed:

$$\begin{cases} \Phi(q^2) = (1\ 2)(1\ 2) = 1 \\ \Phi(q) = (1\ 2) \\ \Phi(pq) = (1\ 3) \\ \Phi(qp) = (2\ 3) \\ \Phi(p) = (1\ 2\ 3) \\ \Phi(pq^2p) = (1\ 3)(2\ 3) = (1\ 3\ 2) \end{cases}$$

and Φ is indeed a morphism.

By contrast, there exists no surjective morphism Ψ from $G(H)$ to $\mathfrak{S}_3$.

Let us argue by contradiction and suppose that such a surjective morphism Ψ exists: necessarily the generators a and d of $G(H)$ are sent onto a generating subset of $\mathfrak{S}_3$. Now the only two-element generating subsets of $\mathfrak{S}_3$ are i) the 6 pairs formed of a transposition and a 3-cycle and ii) the 3 pairs formed of two distinct transpositions. By symmetry of the presentation of $G(H)$ in a and d, for a fixed pair of images we may treat only one of the two permutations of the pair (i.e. a single ordered pair of images).

For each of the 9 possible ordered pairs of images, we show that Ψ is not a morphism. For example, for $a \longmapsto (1\ 2\ 3)$ and $d \longmapsto (1\ 2)$, we have: $\begin{cases} \Psi(ada^{-1}da) = (1\ 2\ 3)(1\ 2)(1\ 3\ 2)(1\ 2)(1\ 2\ 3) \\ \Psi(dad^{-1}ad) = (1\ 2)(1\ 2\ 3)(1\ 2)(1\ 2\ 3)(1\ 2) \\ \Psi(ada^{-1}da) = \Psi(dad^{-1}ad) \end{cases}$

Now $[(1\ 2\ 3)(1\ 2)(1\ 3\ 2)(1\ 2)(1\ 2\ 3)](1) = 1$ and $[(1\ 2)(1\ 2\ 3)(1\ 2)(1\ 2\ 3)(1\ 2)](1) = 2$.

The argument is analogous for the other possible pairs of images.

Finally, the knot groups are not isomorphic.

□

Incompleteness of the knot group invariant:
The knot group is nonetheless not a complete invariant: the square knot $\mathcal{K}_p$ and the granny knot $\mathcal{K}_m$ shown below satisfy $G(\mathcal{K}_p) \cong G(\mathcal{K}_m) \cong \langle a, b, c \mid aba = bab,\ aca = cac \rangle$, and yet they are not isotopic (according to other invariants, notably polynomial ones).



Figure 50: Square knot and granny knot, not isotopic.

2.6 COMPUTATIONAL CONSIDERATIONS

In this section we propose to address a more practical aspect of handling knot invariants and searching for isotopies.

2.6.1 GAUSS CODE

Definition 2.6.1 (Gauss code): The Gauss code of a knot is the sequence of the numbers of its crossings, arbitrarily numbered beforehand, obtained by travelling naturally along the knot, each preceded by the letter U or O indicating whether one passes over or under the crossing.

Example 2.6.2 (Trefoil knot): Here is **one** Gauss code of the trefoil knot: $U1\ O2\ U3\ O1\ U2\ O3$.



Figure: Oriented trefoil knot.

Remark 2.6.3: We shall not distinguish between Gauss codes, words and diagrams.

Definition 2.6.4: A Gauss diagram is said to be realisable if it comes from a knot.

Definition 2.6.5 (Extended Gauss code): The extended Gauss code corresponds exactly to the Gauss code defined above, with the difference that a sign corresponding to the sign of the crossing is appended to the number.

Example 2.6.6 (Trefoil knot):
Here is an extended Gauss word of the trefoil knot: $U1-\ O2-\ U3-\ O1-\ U2-\ O3-$.

2.6.2 ALGORITHMS AND LIMITS

Here we seek to present several existing algorithms on various invariants, and to draw the distinction between accuracy at the cost of complexity, and speed at the cost of correctness.

The first, intuitive idea for telling whether two knots are isotopic is to carry out every possible manipulation of the first one in order to reach the second.
However, since in certain situations a knot has to be made more complicated before it can be simplified, there is a potentially infinite number of manipulations (Reidemeister moves, so to speak) to be carried out.

To answer this problem, we introduce the following theorem of Coward & Lackenby:

Theorem 2.6.7: Consider two isotopic knots K_1 and K_2 having n_1 and n_2 crossings respectively. Set $n = n_1 + n_2$. Then the number of Reidemeister moves needed to pass from K_1 to K_2 is bounded above by: $2^{2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}}$, where the tower of exponentials contains $10^{10^{10^{10^{n}}}}$ 2s.
In what follows, we shall write this number M_{K_1,K_2}.

Thus, the isotopy of two knots can *easily* be checked:

Proposition 2.6.8: Consider two knots K_1 and K_2 having n_1 and n_2 crossings respectively. To check the isotopy of K_1 and K_2, one will have to carry out (in the worst case) $(4n_1)^{M_{K_1,K_2}}$ or $(4n_2)^{M_{K_1,K_2}}$ Reidemeister moves on K_1 or K_2, depending on whether n_1 is smaller than n_2 or not.

Proof: It suffices to see that this amounts to determining (for K_1, say) all the M_{K_1,K_2}-tuples of $\{(c_i, R_j)|(i,j) \in [\![1, n_1]\!], [\![1, 4]\!]\}$. This set itself contains $4n_1$ elements.

Remark 2.6.9: Given that two knots taken at random are *very* rarely isotopic, this number is the average complexity of the naive algorithm proposed.

One quickly realises the futility of this algorithm, even though it theoretically provides a foolproof method for implementing what is indeed a complete invariant.

3 OUTLOOK: FINITE-TYPE INVARIANTS

We have built many invariants, but none of them is complete. The following paragraph sketches, by way of an outlook, Vassiliev's theory of finite-type invariants.

By relaxing the injectivity assumption on the map γ of definition 1.1, we allow ourselves to consider "singular" knots with double points.

Definition 3.1 (n-singular knot): An n-singular knot is the image of $\mathcal{S}^1$ in $\mathbb{R}^3$ under a continuous map γ, having exactly n multiple points, which in this case are double points, written $c_1, ..., c_n$. We shall write Σ_n for the set of n-singular knots.

Setting $\mathfrak{F}$ to be the set of all knots (including singular ones), we have $\mathfrak{F} = \bigsqcup_{n\in\mathbb{N}} \Sigma_n$.

The idea is then to build an invariant adapted to *each pair of knots to be distinguished*. For example, suppose we would like to distinguish the figure-eight knot from the unknot. Here is how to proceed: We set $\varpi_0(\mathcal{O}) = 0$ for the unknot. Then we take a knot M, which moves on the map of strata shown below: it starts from the unknot, then, by the appearance and disappearance of singular points, it is progressively transformed into the figure-eight knot. We stipulate that each time M passes through Σ_1 in the direction of the arrows of the diagram, the value of $\varpi_0(M)$ increases by 1.

ϖ_0 is well defined:

i) The value of ϖ_0 does not depend on the choice of the representative of the knot in its isotopy class

ii) The value of ϖ_0 does not depend on the "path" followed on the strata of $\mathfrak{F}$

iii) ϖ_0 thus built constitutes a knot invariant (insensitivity to the Reidemeister moves)

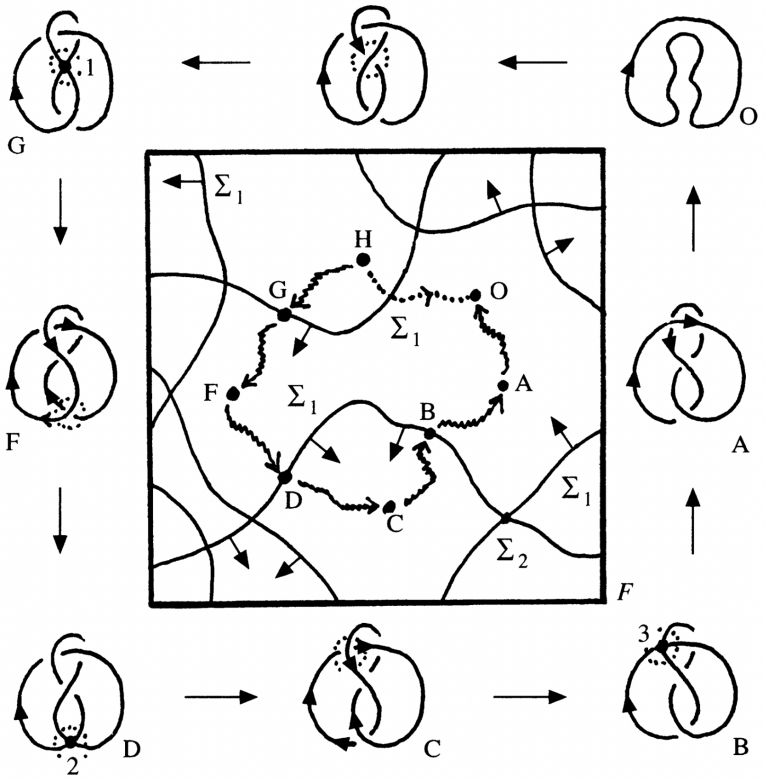


Figure 51: Path of the evolving knot in $\mathfrak{F}$

We imagine the following path:
$\mathcal{O} \to \mathcal{A} \to \mathcal{B} \to \mathcal{C} \to \mathcal{D} \to \mathcal{F} \to \mathcal{G} \to \mathcal{H}$. It then follows that $\varpi_0(\mathcal{H}) = -1 \neq \varpi_0(\mathcal{O})$, which proves that the figure-eight knot is not trivial. This knot invariant therefore manages to distinguish $\mathcal{H}$ and $\mathcal{O}$, but one can show that $\varpi_0(T_1) = \varpi_0(T_2)$ (left and right trefoils). This invariant is therefore incomplete.

Definition 3.2 (Finite-type invariants): An invariant $\Gamma : \mathfrak{F} \longrightarrow E$ of order n is a knot invariant that vanishes on all knots that are at least $n+1$-singular, but not on all n-singular knots. The set $\mathfrak{V}_n$ of invariants of order n is a vector space, and it follows that: $\mathfrak{V}_0 \subset \mathfrak{V}_1 \subset ... \subset \mathfrak{V}_n \subset ...$

We wish to leave our reader with the following conjecture, which has still been neither proved nor disproved, and which nourishes the hope of an "almost" complete invariant:

Conjecture 3.3 (Classification by Vassiliev invariants):
Let K and $\widetilde{K}$ be two non-isotopic knots. Then $\exists n \in \mathbb{N}, \exists \varpi \in \mathfrak{V}_n, \varpi(K) \neq \varpi(\widetilde{K})$.

Let us be clear: this approach is by no means conventional. It runs counter not only to the mathematical formalism of knot theory that has prevailed since the beginning of the twentieth century, but also to the mathematical intuition according to which a knot has no double point (injectivity of γ). Yet this perspective is appealing: with Vassiliev's tools, one can build an invariant *as* powerful as one wishes.

Of course, this takes nothing away from the invariants we have built: tricolourability has the advantage of being visual and quick to determine for simple knots, the polynomial invariants bring us back to algebraic objects that are easy to handle, and the first homotopy group allows a synthesis between the irreducibly topological nature of knots and their algebraic formalisation.

Knot theory therefore has something irremediably dialectical about it: one must constantly change point of view, even if it means returning, as Vassiliev did, to its very foundations. This branch of mathematics does not progress in a single direction: on the contrary, it is open and permeable to all its sister branches, in a certain movement that is mystical, magical and, why not, playful.

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